Differential Equation

89. Let $f:[1,\infty)\to[2,\infty)$ be a differentiable function. If $10\int_1^x f(t)dt = 5xf(x)-x^5-9$ for all $x\ge 1$, then the value of f(3) is :

मानाf : $[1,\infty)\to[2,\infty)$ एक अवकलनीय फलन है, यदि $10\int_1^x f(t)dt = 5xf(x)-x^5-9$, $\forall\ x\ge 1$ तब f(3) का मान होगा:

(A) 18 (B) 22 (C) 26 (D) 32

Ans. D

Sol. $10\int_1^x f(t)dt = 5xf(x)-x^5-9$

differentiate w.r.t. x both sides

$10f(x) = 5f(x) + 5x\,f'(x) - 5x^4$

$\Rightarrow 5f(x) = 5xf'(x) - 5x^4$

Let $y = f(x) \Rightarrow \dfrac{dy}{dx} = f'(x)$

$5y = 5x\dfrac{dy}{dx} - 5x^4 \Rightarrow \dfrac{dy}{dx} - \dfrac{y}{x} = x^3$

I.F. of this linear differential equation

$\Rightarrow \text{I.F.} = e^{\int\frac{-1}{x}dx} = e^{-\ell nx} = e^{\ell n\left(\frac{1}{x}\right)} = \dfrac{1}{x}$

$\Rightarrow y\left(\dfrac{1}{x}\right) = \int x^3\cdot\dfrac{1}{x}dx = \dfrac{x^3}{3}+C$

$\Rightarrow y = \dfrac{x^4}{3}+Cx$, at $x=1$

$10\int_1^1 f(t)dt = 5f(1)-1-9 \Rightarrow f(1)=2$

$\Rightarrow f(1) = \dfrac{1}{3}+C = 2 \Rightarrow C = \dfrac{5}{3}$

$\Rightarrow f(3) = \dfrac{3^4}{3}+\dfrac{5}{3}\cdot 3 = 3^3+5 = 32$

90. Let g be a differentiable function such that $\int_0^x g(t)\,dt = x - \int_0^x tg(t)\,dt$, $x \geq 0$ and let $y = y(x)$ satisfy the differential equation $\frac{dy}{dx} - y\tan x = 2(x+1)\sec x\, g(x)$, $x \in \left[0, \frac{\pi}{2}\right)$. If $y(0) = 0$, then $y\left(\frac{\pi}{3}\right)$ is equal to :

माना एक अवकलनीय फलन g के लिए $\int_0^x g(t)\,dt = x - \int_0^x tg(t)\,dt$, $x \geq 0$ है और माना $y = y(x)$, अवकल समीकरण $\frac{dy}{dx} - y\tan x = 2(x+1)\sec x\, g(x)$, $x \in \left[0, \frac{\pi}{2}\right)$ को संतुष्ट करता है। यदि $y(0) = 0$ है, तो $y\left(\frac{\pi}{3}\right)$ बराबर है :

(A) $\frac{2\pi}{3\sqrt{3}}$ (B) $\frac{4\pi}{3\sqrt{3}}$ (C) $\frac{4\pi}{3}$ (D) $\frac{2\pi}{3}$

Ans. C

Sol. $g(x) = 1 - x\,g(x)$

$$g(x) = \frac{1}{1+x}$$

$$\frac{dy}{dx} - y\tan x = 2\sec x$$

$$\text{I.F.} = e^{-\int \tan x\,dx} = \cos x$$

$$y\cos(x) = \int 2dx + c$$

$$y(0) = 0$$

$$y = \frac{2x}{\cos x}$$

$$y\left(\frac{\pi}{3}\right) = \frac{4\pi}{3}$$

91. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + 3\left(\tan^2 x\right)y + 3y = \sec^2 x$, $y(0) = \frac{1}{3} + e^3$. Then $y\left(\frac{\pi}{4}\right)$ is equal to :

माना अवकल समीकरण $\frac{dy}{dx} + 3\left(\tan^2 x\right)y + 3y = \sec^2 x$, $y(0) = \frac{1}{3} + e^3$ का हल $y = y(x)$ है। तो $y\left(\frac{\pi}{4}\right)$ बराबर है :

(A) $\frac{2}{3}+e^3$ (B) $\frac{4}{3}$ (C) $\frac{4}{3}+e^3$ (D) $\frac{2}{3}$

Ans. B

Sol. $\frac{dy}{dx}+\left(3\sec^2 x\right)y=\sec^2 x$

$IF=e^{3\int\sec^2 x dx}=e^{3\tan x}$

$y\cdot e^{3\tan x}=\int e^{3\tan x}\cdot\sec^2 x dx+c$

$Ye^{3\tan x}=\frac{e^{3\tan x}}{3}+c$

$x=0$ and $y=e^3+\frac{1}{3}$

$e^3+\frac{1}{3}=\frac{1}{3}+C$

$C=e^3$

$ye^{3\tan x}=\frac{e^{3\tan x}}{3}+e^3$

$x=\frac{\pi}{4}$

$ye^3=\frac{e^3}{3}+e^3$

$y=\frac{4}{3}$

92. Let $f:[0,\infty)\to R$ be a differentiable function such that $f(x)=1-2x+\int_0^x e^{x-t}f(t)dt$ for all $x\in[0,\infty)$.

Then the area of the region bounded by y = f(x) and the coordinate axes is :

माना एक अवकलनीय फलन $f:[0,\infty)\to R$ के लिए $f(x)=1-2x+\int_0^x e^{x-t}f(t)dt$, $x\in[0,\infty)$ है। तो y = f(x) और निर्देशांक अक्षों से परिबद्ध क्षेत्र का क्षेत्रफल है :

(A) $\sqrt{2}$ (B) 2 (C) $\sqrt{5}$ (D) $\frac{1}{2}$

Ans. D

Sol. $f:[0,\infty)\to R$

$$f(x) = 1 - 2x + \int_0^x e^{x-t} f(t)\,dt \qquad ...(1) \qquad \forall\, x \in [0, \infty)$$

$$= 1 - 2x + e^x \int_0^x e^{-t} f(t)\,dt$$

$$f'(x) = -2x + e^x \int_0^x e^{-t} f(t)\,dt + e^x e^{-x} f(x)$$

$$= -2 + f(x) - 1 + 2x + f(x)$$

$$\frac{dy}{dx} = -3 + 2y + 2x$$

$$\frac{dy}{dx} - 2y = 2x - 3$$

$P = -2;\ Q = 2x - 3$

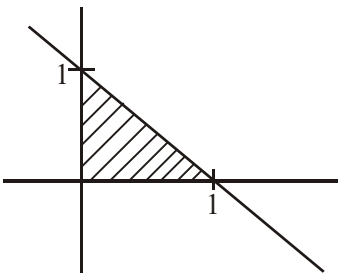
I.F. $= e^{-2x}$

$$y.e^{-2x} = \int e^{-2x}(2x - 3)\,dx$$

$$y \cdot e^{-2x} = \frac{e^{-2x}}{-2}(2x - 3) - 2 \cdot \frac{e^{-2x}}{4} + C$$

$$y = \frac{3 - 2x}{2} - \frac{1}{2} + C \cdot e^{2x}$$

Now, $f(0) = 1$



$$1 = \frac{3}{2} - \frac{1}{2} + C$$

$C = 0$

$$y = \frac{3 - 2x - 1}{2}$$

$$= \frac{2 - 2x}{2}$$

$y = 1 - x$

$$A = \frac{1}{2} \times 1 \times 1$$

$$= \frac{1}{2}$$

93. If a curve y = y(x) passes through the point $\left(1,\frac{\pi}{2}\right)$ and satisfies the differential equation $\left(7x^4\cot y - e^x \operatorname{cosec} y\right)\frac{dx}{dy} = x^5, x \geq 1$, then at x = 2, the value of cosy is :

यदि एक वक्र y = y(x) बिन्दु $\left(1,\frac{\pi}{2}\right)$ से होकर जाता है और अवकल समीकरण $\left(7x^4\cot y - e^x \operatorname{cosec} y\right)\frac{dx}{dy} = x^5, x \geq 1$, को सन्तुष्ट करता है, तो x = 2 पर cosy का मान है :

(A) $\frac{2e^2 - e}{64}$ (B) $\frac{2e^2 + e}{128}$ (C) $\frac{2e^2 - e}{128}$ (D) $\frac{2e^2 + e}{64}$

Ans. C

Sol. $\left(7x^4\cot y - e^x \operatorname{cosec} y\right)\frac{dx}{dy} = x^5$

$7x^4\cos y - e^x = x^5 \sin y \frac{dy}{dx}$

$\cos y = t$

$\frac{7}{x}t - \frac{e^x}{x^5} = \frac{-dt}{dx}$

$\frac{dt}{dx} + \frac{7}{x}t = \frac{e^x}{x^5}$

$I.F. = e^{\int \frac{7}{x}dx} = c^{7\ell nx} = x^7$

$tx^7 = \int e^x . x^2 dx + c$

$\cos y\, x^7 = x^2 e^x - 2x e^x + 2e^x + c$

$\Rightarrow c = -e$

$At\ x = 2 \Rightarrow \cos y = \frac{2e^2 - e}{128}$

94. Let y = y(x) be the solution curve of the differential equation $x(x^2 + e^x)\,dy + (e^x(x-2)y - x^3)\,dx = 0, x > 0$, passing through the point (1, 0). Then y(2) is equal to :

माना अवकल समीकरण $x(x^2 + e^x)\,dy + (e^x(x-2)y - x^3)\,dx = 0, x > 0$ का हल वक्र y = y(x), बिन्दु (1, 0) से होकर जाता है। तो y(2) बराबर है :

(A) $\frac{2}{2-e^2}$ (B) $\frac{4}{4+e^2}$ (C) $\frac{4}{4-e^2}$ (D) $\frac{2}{2+e^2}$

Ans. B

Sol. $x(x^2+e^x)dy+(e^x(x-2)y-x^3)dx=0$

$$x\left(x^2+e^x\right)\frac{dy}{dx}+e^x\left(x-2\right)y=x^3$$

$$\frac{dy}{dx}+\frac{e^x\left(x-2\right)}{x\left(x^2+e^x\right)}y=\frac{x^2}{x^2+e^x}$$

$$\text{I.F.}=e^{\int\frac{e^x(x-2)}{x\left(x^2+e^x\right)}dx}=e^{\int\frac{e^x\left(\frac{1}{x^2}-\frac{2}{x^2}\right)}{\left(1+\frac{e^x}{x^2}\right)}dx}$$

$$\text{Let } 1+\frac{e^x}{x^2}=t\Rightarrow\frac{x^2e^x-e^x2x}{x^4}dx=dt$$

$$\Rightarrow \text{I.F. } e^{\ell n\left(1+\frac{e^x}{x^2}\right)}=1+\frac{e^x}{x^2}$$

$$\text{Now } y\left(1+\frac{e^x}{x^2}\right)=\int\frac{x^2}{x^2+e^x}\cdot\frac{x^2+e^x}{x^2}dx+C$$

$$y\left(1+\frac{e^x}{x^2}\right)=x+C$$

Passing through (1, 0)

$\Rightarrow C=-1$

$$y=\frac{x-1}{1+\frac{e^x}{x^2}}$$

$$y(2)=\frac{1}{1+\frac{e^2}{4}}=\frac{4}{4+e^2}$$

95. Let $y=y(x)$ be the solution of the differential equation $(x^2+1)y'-2xy=(x^4+2x^2+1)\cos x$, $y(0)=1$. Then $\int_{-3}^{3}y(x)dx$ is :

माना अवकल समीकरण $(x^2 + 1)y' - 2xy = (x^4 + 2x^2 + 1)\cos x$, $y(0) = 1$ का हल $y = y(x)$ है। तो $\int_{-3}^{3} y(x)\,dx$ का मान है :

(A) 36 (B) 18 (C) 24 (D) 30

Ans. C

Sol. $\left(x^2+1\right)\frac{dy}{dx} - 2xy = \left(x^2+1\right)^2 \cos x$

$\Rightarrow \frac{1}{\left(x^2+1\right)}\frac{dy}{dx} - \frac{2x}{\left(x^2+1\right)^2}\cdot y = \cos x$

$\Rightarrow \frac{d}{dx}\left(y\cdot\frac{1}{1+x^2}\right) = \cos x$

$\Rightarrow \frac{y}{1+x^2} = \sin x + c$

$y(0) = 1$

$\Rightarrow 1 = c$

So, $y = (\sin x + 1)(x^2 + 1)$

$\int_{-3}^{3} y(x)\,dx = \int_{-3}^{3} \sin x\cdot\left(x^2+1\right)dx + \int_{-3}^{3}\left(x^2+1\right)dx$

$= 0 + 2\left(\frac{x^3}{3} + x\right)_0^3$

$= 2 \times 12$

$= 24$

96. Let $f(x) = x - 1$ and $g(x) = e^x$ for $x \in R$. If $\frac{dy}{dx} = \left(e^{-2\sqrt{x}} g\left(f\left(f(x)\right)\right) - \frac{y}{\sqrt{x}}\right)$, $y(0) = 0$, then $y(1)$ is :

माना $f(x) = x - 1$ और $g(x) = e^x$, $x \in R$ हैं। यदि $\frac{dy}{dx} = \left(e^{-2\sqrt{x}} g\left(f\left(f(x)\right)\right) - \frac{y}{\sqrt{x}}\right)$, $y(0) = 0$ हैं, तो $y(1)$ बराबर है :

(A) $\frac{1-e^2}{e^4}$ (B) $\frac{2e-1}{e^3}$ (C) $\frac{e-1}{e^4}$ (D) $\frac{1-e^3}{e^4}$

Ans. C

Sol. $P(f(x)) = (x-1) - 1 = x - 2$

$g(ff(x)) = ex^{-2}$

Now $\frac{dy}{dx} = e^{-2\sqrt{x}}.e^{x-2} - \frac{1}{\sqrt{x}}y$

$\frac{dy}{dx} + \frac{1}{\sqrt{x}}.y = e^{-2\sqrt{x}+x-2}$ _____(1)

I.F. $= e^{\int \frac{dx}{\sqrt{x}}} = e^{2\sqrt{x}}$

Solving of equation : $y.e^{2\sqrt{x}} = \int e^{x-2}dx$

$y.e^{2\sqrt{x}} = e^{x-2} + C$

$y.e^{2\sqrt{x}} = e^{x-2} - e^{-2}$ $\quad (\because y(0) = 0)$

$y(1) = \frac{e^{-1} - e^{-2}}{e^2} = \frac{e-1}{e^4}$

97. Let x = x(y) be the solution of the differential equation $y^2\,dx + \left(x - \frac{1}{y}\right)dy = 0$. If x(1) = 1, then $x\left(\frac{1}{2}\right)$ is :

माना कि x = x(y), अवकल समीकरण $y^2\,dx + \left(x - \frac{1}{y}\right)dy = 0$ का हल है। यदि x(1) = 1 है, तो $x\left(\frac{1}{2}\right)$ है :

(A) 3 – e (B) $\frac{1}{2} + e$ (C) $\frac{3}{2} + e$ (D) 3 + e

Ans. A

Sol. $\frac{dx}{dy} = \frac{-x}{y^2} + \frac{1}{y^3}$

$\frac{dx}{dy} + \frac{x}{y^2} = \frac{1}{y^3}$

I.F. $= e^{\int \frac{1}{y^2}dy} = e^{-\frac{1}{y}}$

Now, solution of differential equation

$x.e^{-\frac{1}{y}} = \int e^{-\frac{1}{y}} \cdot \frac{1}{y^3}dy + C$

Put $-\frac{1}{y} = t$

$$\frac{1}{y^2}dy = dt$$

$$xe^{-\frac{1}{y}} = -\int e^t.tdt + C$$

$$xe^{-\frac{1}{y}} = -\left[te^t - e^t\right] + C$$

$$xe^{-\frac{1}{y}} = e^t(1-t) + C$$

$$xe^{-\frac{1}{y}} = e^{-\frac{1}{y}}\left(1+\frac{1}{y}\right) + C$$

Now, when $y = 1 \Rightarrow x = 1$

We get, $1.e^{-1} = e^{-1}(1 + 1) + C$

$$C = -\frac{1}{e}$$

Now, $xe^{-\frac{1}{y}} = e^{\frac{-1}{y}}\left(1+\frac{1}{y}\right) - \frac{1}{e}$

Put $y = \frac{1}{2}$, we get

$x.e^{-2} = e^{-2}(1 + 2) – e^{-1}$

$x = 3 – e$

98. Let f(x) be a real differentiable function such that f(0) = 1 and f(x + y) = f(x) f'(y) + f'(x) f(y) for all x, y ∈ R. Then $\sum_{n=1}^{100} \log_e f(n)$ is equal to :

माना कि f(x) एक वास्तविक अवकलनीय फलन इस प्रकार है कि f(0) = 1 है और सभी x, y ∈ R के लिए f(x + y) = f(x) f'(y) + f'(x) f(y) है। तब $\sum_{n=1}^{100} \log_e f(n)$ बराबर है :

(A) 2406 (B) 5220 (C) 2525 (D) 2384

Ans. C

Sol. Put x = y = 0

f(0) = 2f′(0) as f(0) = 1

$f'(0)=\frac{1}{2}$

Now, put y = 0 in given equation

f(x) = f(x).f′(0) + f(0).f′(x)

$f(x)=\frac{1}{2}f(x)+f'(x)$

$f'(x)=\frac{1}{2}f(x)$

$\frac{dy}{dx}=\frac{y}{2}$

$\int\frac{dy}{y}=\int\frac{dx}{2}$

$\ell ny=\frac{x}{2}+C$

f(0) = 1

0 = 0 + C

C = 0

$y=e^{\frac{x}{2}}$

$\ell ny=\frac{x}{2}$

Now, $\sum_{n=1}^{100}\ell nf(n)=\frac{1}{2}+\frac{2}{2}+\frac{3}{2}+.....\frac{100}{2}=\frac{1}{2}\left[\frac{100\times101}{2}\right]=2525$

99. If x = f(y) is the solution of the differential equation $\left(1+y^2\right)+\left(x-2e^{\tan^{-1}y}\right)\frac{dy}{dx}=0$, $y\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ with f(0) = 1, then $f\left(\frac{1}{\sqrt{3}}\right)$ is equal to :

यदि x = f(y), f(0) = 1 के साथ अवकल समीकरण $\left(1+y^2\right)+\left(x-2e^{\tan^{-1}y}\right)\frac{dy}{dx}=0$, $y\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ का हल है, तो $f\left(\frac{1}{\sqrt{3}}\right)$ बराबर है :

(A) $e^{\frac{\pi}{6}}$ (B) $e^{\frac{\pi}{4}}$ (C) $e^{\frac{\pi}{12}}$ (D) $e^{\frac{\pi}{3}}$

Ans. A

Sol. $x = f(y)$

$$\left(1+y^2\right)+\left(x-2e^{\tan^{-1}y}\right)\frac{dy}{dx}=0$$

$$\frac{dx}{dy}+\underbrace{\frac{1}{1+y^2}}_{P}x=\underbrace{\frac{2e^{\tan^{-1}y}}{1+y^2}}_{Q}$$

$$\text{I.F.} = e^{\int p dy} = e^{\int \frac{1}{1+y^2}dy} = e^{\tan^{-1}y}$$

Now $$x\times e^{\tan^{-1}y} = \int \frac{2e^{\tan^{-1}y}}{1+y^2}\times e^{\tan^{-1}y}dy + C$$

$$= \int \frac{2\left(e^{\tan^{-1}y}\right)^2}{1+y^2}dy + C$$

$$xe^{\tan^{-1}y} = \int \frac{2e^{2\tan^{-1}y}}{1+y^2}dy + C$$

put $e^{2\tan^{-1}y} = z$

$$e^{2\tan^{-1}y}\times\frac{2}{1+y^2}dy = dz$$

$$xe^{\tan^{-1}y} = \int 1dz + C$$

$$xe^{\tan^{-1}y} = z + c$$

$$xe^{\tan^{-1}y} = e^{2\tan^{-1}y} + C$$

$$x = e^{\tan^{-1}y} + \frac{C}{e^{\tan^{-1}y}}$$

given $f(0) = 1$

$y = 0, \quad x = 1$

$$1 = e^0 + \frac{C}{e^0}$$

$C = 0$

$$x = e^{\tan^{-1}y}$$

Now put $y = \frac{1}{\sqrt{3}}$

$x = e^{\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)} = e^{\frac{\pi}{6}}$

100. Let x = x(y) be the solution of the differential equation $y = \left(x - y\frac{dx}{dy}\right)\sin\left(\frac{x}{y}\right), y > 0$ and $x(1) = \frac{\pi}{2}$. Then cos (x(2)) is equal to :

माना कि x = x(y) अवकल समीकरण $y = \left(x - y\frac{dx}{dy}\right)\sin\left(\frac{x}{y}\right), y > 0$ का हल है और $x(1) = \frac{\pi}{2}$ है। तब cos (x(2)) बराबर है :

(A) $2(\log_e 2)^2 - 1$ (B) $2(\log_e 2) - 1$ (C) $1 - 2(\log_e 2)$ (D) $1 - 2(\log_e 2)^2$

Ans. A

Sol. $y = \left(x - y \cdot \frac{dx}{dy}\right) \cdot \sin\left(\frac{x}{y}\right)$

$y\,dy = (x\,dy - y\,dx) \cdot \sin\frac{x}{y}$

$\frac{y\,dy}{y^2} = \frac{(x\,dy - y\,dx)}{y^2} \cdot \sin\left(\frac{x}{y}\right)$

$\int \frac{dy}{y} = \int \sin\left(\frac{x}{y}\right) \cdot d\left(\frac{x}{y}\right)$

$\ell n y = -\cos\frac{x}{y} + c$

$y = 1, \quad x = \frac{\pi}{2}$

$O = -\cos\frac{\pi}{2} + c$

$c = 0$

$\therefore \ell n y = -\cos\frac{x}{y}$

for $y = 2$

$\ell n2 = -\cos\frac{x}{2}$

$\cos\frac{x}{2} = -\ell n2$

$\cos x = 2\cos^2\frac{x}{2} - 1$

$= 2\ell n^2 2 - 1$

101. Let y = y(x) be the solution of the differential equation $\left(xy - 5x^2\sqrt{1+x^2}\right)dx + \left(1+x^2\right)dy = 0, y(0) = 0$. Then $y\left(\sqrt{3}\right)$ is equal to :

माना कि y = y(x) अवकल समीकरण $\left(xy - 5x^2\sqrt{1+x^2}\right)dx + \left(1+x^2\right)dy = 0, y(0) = 0$ का हल है। तब $y\left(\sqrt{3}\right)$ बराबर है :

(A) $2\sqrt{2}$ (B) $\sqrt{\frac{14}{3}}$ (C) $\frac{5\sqrt{3}}{2}$ (D) $\sqrt{\frac{15}{2}}$

Ans. C

Sol. $\text{I.F.} = \int\frac{xdx}{1+x^2} = e^{\frac{1}{2}\ell n\left(1+x^2\right)} = \sqrt{1+x^2}$

Solution of differential equation

$y.\sqrt{1+x^2} = \int\sqrt{1+x^2}\cdot\frac{5x^2}{\sqrt{1+x^2}}dx + C$

$y\sqrt{1+x^2} = \frac{5x^3}{3} + C$

when x = 0, y = 0 then C = 0

So, $y = \frac{5x^3}{3\sqrt{1+x^2}}$

$\therefore y\left(\sqrt{3}\right) = \frac{5.3\sqrt{3}}{3.2} = \frac{5\sqrt{3}}{2}$

102. Let for some function y = f(x), $\int_0^x tf\left(t\right)dt = x^2 f\left(x\right), x > 0$ and f(2) = 3. Then f(6) is equal to :

माना कि किसी फलन $y = f(x)$ के लिए, $\int_0^x tf(t)dt = x^2 f(x)$, $x > 0$ और $f(2) = 3$ है। तब $f(6)$ बराबर है :

(A) 1 (B) 3 (C) 6 (D) 2

Ans. A

Sol. $\int_0^x tf(t)dt = x^2 f(x)$, $x > 0$

Differentiating both side w.r.t. x

$xf(x) = x^2 f'(x) + 2xf(x)$

$-xf(x) = x^2 f'(x)$

$$\int \frac{f'(x)}{f(x)}dx = \int \frac{-1}{2}dx$$

$\log df(x) = -\log x + \log c$

$$f(x) = \frac{c}{x}$$

$$f(2) = 3 \Rightarrow 3 = \frac{c}{2} \Rightarrow c = 6$$

$$f(x) = \frac{6}{x}$$

$f(6) = 1$ $\therefore$ (1)

103. Let $y = y(x)$ be the solution of the differential equation

$\cos x\left(\log_e(\cos x)\right)^2 dy + \left(\sin x - 3y\sin x \log_e(\cos x)\right)dx = 0$, $x \in \left(0, \frac{\pi}{2}\right)$. If $y\left(\frac{\pi}{4}\right) = \frac{-1}{\log_e 2}$, then $y\left(\frac{\pi}{6}\right)$ is equal to :

माना कि $y = y(x)$, अवकल समीकरण $\cos x\left(\log_e(\cos x)\right)^2 dy + \left(\sin x - 3y\sin x \log_e(\cos x)\right)dx = 0$, $x \in \left(0, \frac{\pi}{2}\right)$ का हल है। यदि $y\left(\frac{\pi}{4}\right) = \frac{-1}{\log_e 2}$ है, तो $y\left(\frac{\pi}{6}\right)$ बराबर है :

(A) $-\frac{1}{\log_e(4)}$ (B) $\frac{2}{\log_e(3) - \log_e(4)}$ (C) $\frac{1}{\log_e(4) - \log_e(3)}$ (D) $\frac{1}{\log_e(3) - \log_e(4)}$

Ans. D

Sol. $\cos x(\ell n \cos x)^2 dy = (3y \sin x \ell n \cos x - \sin x)dx$

$$\frac{dy}{dx}=\frac{3y\sin x\ell n\cos x}{\cos x\left(\ell n\cos x\right)^2}-\frac{\tan x}{\left(\ell n\cos x\right)^2}$$

$$\frac{dy}{dx}-\frac{3\tan x}{\ell n\cos x}=-\frac{\tan x}{\left(\ell n\cos x\right)^2}$$

$$\text{I.F.}=e^{\int-\frac{3\tan x}{\ell n\cos x}dx}$$

$$\ell n\cos x=t$$

$$\frac{1}{\cos x}\times-\sin x dx=dt$$

$$-\tan x\,dx=dt$$

$$\text{I.F.}=e^{\int\frac{3dt}{t}}=e^{3\ell nt}=t^3=\left(\ell n\cos x\right)^3$$

$$y\times\left(\ell n\cos x\right)^3=-\int\tan x\left(\ell n\cos x\right)dx$$

$$\ell n\cos x=t$$

$$-\tan x\,dx=dt$$

$$y\left(\ell n\cos x\right)^3=\frac{t^2}{2}+C$$

$$y\left(\ell n\cos x\right)^3=\frac{\left(\ell n\cos x\right)^2}{2}+C$$

$$C=0$$

$$y=\frac{1}{2\ell n\cos x}$$

$$y=\frac{1}{2\ell n\frac{\sqrt{3}}{2}}$$

$$y=\frac{1}{2\left(\ell n\sqrt{3}-\ell n2\right)}$$

$$y=\frac{1}{\ell n3-\ell n4}$$

104. If for solution curve $y=f(x)$ of the differential equation $\frac{dy}{dx}+\left(\tan x\right)y=\frac{2+\sec x}{\left(1+2\sec x\right)^2}$, $x\in\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$, $f\left(\frac{\pi}{3}\right)=\frac{\sqrt{3}}{10}$,

then $f\left(\frac{\pi}{4}\right)$ is equal to :

यदि अकवल समीकरण $\frac{dy}{dx}+(\tan x)y=\frac{2+\sec x}{(1+2\sec x)^2}$, $x\in\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$ के हल वक्र $y=f(x)$ के लिए $f\left(\frac{\pi}{3}\right)=\frac{\sqrt{3}}{10}$ है, तो $f\left(\frac{\pi}{4}\right)$ बराबर है :

(A) $\frac{4-\sqrt{2}}{14}$ (B) $\frac{5-\sqrt{3}}{2\sqrt{2}}$ (C) $\frac{9\sqrt{3}+3}{10\left(4+\sqrt{3}\right)}$ (D) $\frac{\sqrt{3}+1}{10\left(4+\sqrt{3}\right)}$

Ans. B

Sol. If $e^{\int\tan dx x}=e^{\ell n(\sec x)}=\sec x$

$$\therefore y\cdot\sec x=\int\left\{\frac{2+\sec x}{(1+2\sec x)^2}\right\}\sec x dx$$

$$=\int\frac{2\cos x+1}{(\cos x+2)^2}dx$$

Let $\cos x=\frac{1-t^2}{1+t^2}$

$$=\int\frac{2\left(\frac{1-t^2}{1+t^2}\right)+1}{\left(\frac{1-t^2}{1+t^2}+2\right)^2}2dt$$

$$=\int\frac{2-2t^2+1+t^2}{\left(1-t^2+2+2t^2\right)^2}\times 2dt$$

$$=2\int\frac{3-t^2}{\left(t^2+3\right)^2}dt$$

Let $t+\frac{3}{t}=u$

$$\left(1-\frac{3}{t^2}\right)dt=du$$

$= -2\int \frac{du}{u^2}$

$y \cdot (\sec x) = \frac{2}{u} + c$

$y \cdot \sec x = \frac{2}{t + \frac{3}{t}} + c$ _____(i)

At $x = \frac{\pi}{3}$, $t = \tan\frac{x}{2} = \frac{1}{\sqrt{3}}$

$2.\frac{\sqrt{3}}{10} = \frac{2}{\frac{1}{\sqrt{3}} + 3\sqrt{3}} + c$

$2.\frac{\sqrt{3}}{10} = \frac{2\sqrt{3}}{10} + c \Rightarrow C = 0$

At $x = \frac{\pi}{4}$, $t = \tan\frac{x}{2} = \sqrt{2} - 1$

$\therefore y \cdot \sqrt{2} = \frac{2}{\sqrt{2} - 1 + \frac{3}{\sqrt{2} - 1}}$

$y \cdot \sqrt{2} = \frac{2(\sqrt{2} - 1)}{6 - 2\sqrt{2}}$

$y = \frac{\sqrt{2}(\sqrt{2} - 1)}{2(3 - \sqrt{2})} = \frac{1}{\sqrt{2}} \times \frac{2\sqrt{2} - 1}{7}$

$= \frac{4 - \sqrt{2}}{14}$

105. If the solution $y = y(x)$ of the differential equation $(x^4 + 2x^3 + 3x^2 + 2x + 2)dy - (2x^2 + 2x + 3)dx = 0$ satisfies $y(-1) = -\frac{\pi}{4}$, then $y(0)$ is equal to :

यदि अवकल समीकरण $(x^4 + 2x^3 + 3x^2 + 2x + 2)dy - (2x^2 + 2x + 3)dx = 0$ का हल $y = y(x)$, $y(-1) = -\frac{\pi}{4}$ को संतुष्ट करता है, तो $y(0)$ बराबर है :

(A) 0 (B) $-\frac{\pi}{12}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

Ans. D

Sol. $\int dy = \int \frac{\left(2x^2+2x+3\right)}{x^4+2x^3+2x+2}dx$

$y = \int \frac{\left(2x^2+2x+3\right)}{\left(x^2+1\right)\left(x^2+2x+2\right)}dx$

$y = \int \frac{dx}{x^2+2x+2} + \int \frac{dx}{x^2+1}$

$y = \tan^{-1}(x+1) + \tan^{-1}x + C$

$y(-1) = -\frac{\pi}{4}$

$-\frac{\pi}{4} = 0 - \frac{\pi}{4} + C \qquad \Rightarrow C = 0$

$\Rightarrow y = \tan^{-1}(x+1) + \tan^{-1}x$

$y(0) = \tan^{-1}1 = \frac{\pi}{4}$

106. Let $y = y(x)$ be the solution of the differential equation $(x^2+4)^2dy + (2x^3y + 8xy - 2)dx = 0$. If $y(0) = 0$, then $y(2)$ is equal to :

माना $y = y(x)$ अवकल समीकरण $(x^2+4)^2dy + (2x^3y + 8xy - 2)dx = 0$ का हल है। यदि $y(0) = 0$ है, तो $y(2)$ बराबर है:

(A) $\frac{\pi}{32}$ (B) $\frac{\pi}{16}$ (C) 2π (D) $\frac{\pi}{8}$

Ans. A

Sol. $\frac{dy}{dx} + \left(\frac{2x}{x^2+4}\right)y = \frac{2}{\left(x^2+4\right)^2}$

I.F. $= e^{\int \frac{2x}{x^2+4}dx} = e^{\ln\left(x^2+4\right)} = x^2+4$

solution of differential equation $y\left(x^2+4\right) = \int \frac{2}{x^2+4}dx \Rightarrow y\left(x^2+4\right) = \tan^{-1}\left(\frac{x}{2}\right) + C$

But $y(0) = 0 \therefore c = 0$

$$y\left(x^2+4\right)=\tan^{-1}\left(\frac{x}{2}\right)$$

$$\text{at } x=2 \rightarrow y=\frac{\pi/4}{8}=\frac{\pi}{32}$$

107. The differential equation of the family of circles passing through the origin and having center at the line y = x is:

मूल बिन्दु से होकर जाने वाले तथा रेखा y = x पर केन्द्र के वृत्त कुल का अवकल समीकरण है :

(A) $(x^2-y^2+2xy)dx=(x^2-y^2-2xy)dy$ (B) $(x^2+y^2+2xy)dx=(x^2-y^2+2xy)dy$

(C) $(x^2+y^2-2xy)dx=(x^2+y^2+2xy)dy$ (D) $(x^2+y^2+2xy)dx=(x^2+y^2-2xy)dy$

Ans. A

Sol. $C \equiv x^2+y^2+gx+gy=0$ .....(1)

$2x+2yy'+g+gy'=0$

$$g=-\left(\frac{2x+2yy'}{1+y'}\right)$$

put in (1)

$$x^2+y^2-\left(\frac{2x+2yy'}{1+y'}\right)(x+y)=0$$

$$\left(x^2-y^2-2xy\right)y'=x^2-y^2+2xy$$

108. Let y = y(x) be the solution of the differential equation $\left(2x\log_e x\right)\frac{dy}{dx}+2y=\frac{3}{x}\log_e x, x>0$ and $y\left(e^{-1}\right)=0$. Then, y(e) is equal to :

माना y = y(x) अवकल समीकरण $\left(2x\log_e x\right)\frac{dy}{dx}+2y=\frac{3}{x}\log_e x, x>0$ तथा $y\left(e^{-1}\right)=0$ का हल है, तो y(e) बराबर है:

(A) $-\frac{3}{2e}$ (B) $-\frac{3}{e}$ (C) $-\frac{2}{e}$ (D) $-\frac{2}{3e}$

Ans. B

Sol. $\left(1+x^2\right)\cdot\frac{dy}{dx}+y=e^{\tan^{-1}x}$

Let $\tan^{-1}x=t$

$$\frac{1}{1+x^2}\cdot dx=dt$$

$$\frac{dy}{dt}+y=e^t$$

$$\text{If}=e^{\int 1\cdot dt}=e^t$$

$$y\cdot e^t=\frac{e^{2t}}{2}+C$$

$$y\cdot e^{\tan^{-1}x}=\frac{e^{2\tan^{-1}x}}{2}+C$$

$$x=1 \quad y=0 \quad C=-\frac{e^{\pi/2}}{2}$$

$$y\cdot \tan^{-1}x=\frac{e^{2\tan^{-1}x}}{2}-\frac{e^{\pi/2}}{2}$$

$$x=0\Rightarrow y=\frac{1}{2}\left(1-e^{\pi/2}\right)$$

109. Let y = y(x) be the solution of the differential equation $\left(1+x^2\right)\frac{dy}{dx}+y=e^{\tan^{-1}x}$, y(1) = 0. Then y(0) is :

माना y = y(x) अवकल समीकरण $\left(1+x^2\right)\frac{dy}{dx}+y=e^{\tan^{-1}x}$, y(1) = 0 का हल है, तो y(0) है :

(A) $\frac{1}{2}\left(1-e^{\pi/2}\right)$ (B) $\frac{1}{4}\left(e^{\pi/2}-1\right)$ (C) $\frac{1}{4}\left(1-e^{\pi/2}\right)$ (D) $\frac{1}{2}\left(e^{\pi/2}-1\right)$

Ans. A

Sol. $\left(1+x^2\right)\cdot\frac{dy}{dx}+y=e^{\tan^{-1}x}$

Let $\tan^{-1}x=t$

$$\frac{1}{1+x^2}\cdot dx=dt$$

$$\frac{dy}{dt}+y=e^t$$

$$\text{If}=e^{\int 1\cdot dt}=e^t$$

$$y\cdot e^t=\frac{e^{2t}}{2}+C$$

$$y \cdot e^{\tan^{-1} x} = \frac{e^{2\tan^{-1} x}}{2} + C$$

$$x = 1 \quad y = 0 \quad C = -\frac{e^{\pi/2}}{2}$$

$$y \cdot \tan^{-1} x = \frac{e^{2\tan^{-1} x}}{2} - \frac{e^{\pi/2}}{2}$$

$$x = 0 \quad \Rightarrow y = \frac{1}{2}\left(1 - e^{\pi/2}\right)$$

110. Suppose the solution of the differential equation $\frac{dy}{dx} = \frac{(2+\alpha)x - \beta y + 2}{\beta x - 2\alpha y - (\beta\gamma - 4\alpha)}$ represents a circle passing through origin. Then the radius of this circle is :

मान लीजिए कि अवकल समीकरण $\frac{dy}{dx} = \frac{(2+\alpha)x - \beta y + 2}{\beta x - 2\alpha y - (\beta\gamma - 4\alpha)}$ का हल मूल बिन्दु से गुजरने वाले वृत्त को दर्शाता है, तो इस वृत्त की त्रिज्या है :

(A) $\frac{\sqrt{17}}{2}$ (B) 2 (C) $\sqrt{17}$ (D) $\frac{1}{2}$

Ans. A

Sol. $$\frac{dy}{dx} = \frac{(2+\alpha)x - \beta y + 2}{\beta x - y(2\alpha+\beta) + 4\alpha}$$

$$\beta x dy - (2\alpha+\beta) y dy + 4\alpha dy = (2+\alpha) x dx - \beta y dx + 2dx$$

$$\beta(x dy + y dx) - (2\alpha+\beta) y dy + 4\alpha dy = (2+\alpha) x dx + 2dx$$

$$\beta xy - \frac{(2\alpha+\beta)y^2}{2} + 4\alpha y = \frac{(2+\alpha)x^2}{2}$$

$\Rightarrow \beta = 0$ for this to be circle

$$(2+\alpha)\frac{x^2}{2} + \alpha y^2 + 2x - 4\alpha y = 0$$

coeff. of $x^2 = y^2$ $\rightarrow 2 + a = 2a$

$\Rightarrow \alpha = 2$

i.e. $2x^2 + 2y^2 + 2x - 8y = 0$

$x^2 + y^2 + x - 4y = 0$

$$rd = \sqrt{\frac{1}{4}+4} = \frac{\sqrt{17}}{2}$$

111. Let $y = y(x)$ be the solution of the differential equation $(1 + y^2)e^{\tan x}\,dx + \cos^2 x\,(1 + e^{2\tan x})dy = 0$, $y(0) = 1$. Then $y\left(\frac{\pi}{4}\right)$ is equal to :

माना $y = y(x)$ अवकल समीकरण $(1 + y^2)e^{\tan x}\,dx + \cos^2 x\,(1 + e^{2\tan x})dy = 0$ का हल है एवं $y(0) = 1$ है तब $y\left(\frac{\pi}{4}\right)$ बराबर होगा:

(A) $\frac{2}{e}$ (B) $\frac{2}{e^2}$ (C) $\frac{1}{e^2}$ (D) $\frac{1}{e}$

Ans. D

Sol. $(1 + y^2)e^{\tan x}\,dx + \cos^2 x\,(1 + e^{2\tan x})dy = 0$

$$\frac{dy}{1+y^2} = -\frac{e^{\tan x}dx}{\cos^2 x\left(1+e^{2\tan x}\right)}$$

Let $e^{\tan x} = t$

then $e^{\tan x} \cdot \sec^2 x\,dx = dt$

$$\Rightarrow \tan^{-1} y = -\frac{dt}{\left(1+t^2\right)}$$

$\Rightarrow \tan^{-1}y = -\tan^{-1}(t) + C$

$\Rightarrow \tan^{-1}y + \tan^{-1}(e^{\tan x}) = C$

$y(0) = 1$

$\Rightarrow \tan^{-1}(1) + \tan^{-1}(1) = C$

$$C = \frac{\pi}{2}$$

$$\rightarrow \tan^{-1} y + \tan^{-1}\left(e^{\tan x}\right) = \frac{\pi}{2}$$

$$\rightarrow y\left(\frac{\pi}{4}\right) = ?$$

$$\Rightarrow \tan^{-1}(y) + \tan^{-1}(e) = \frac{\pi}{2}$$

$$\Rightarrow \tan^{-1}\left(\frac{y+e}{1-ye}\right)=\frac{\pi}{2}$$

$$\frac{y+e}{1-ye}=\tan\left(\frac{\pi}{2}\right)$$

$$y=\frac{1}{e}$$

112. Let f(x) be a positive function such that the area bounded by y = f(x), y = 0 from x = 0 to x = a > 0 is $e^{-a} + 4a^2 + a - 1$. Then the diffrential equation, whose general solution is $y = c_1 f(x) + c_2$, where c_1 and c_2 are arbitrary constants, is :

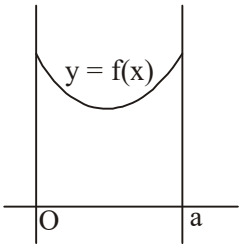
माना f(x) एक धनात्मक फलन इस प्रकार है कि y = f(x), y = 0 द्वारा x = 0 से x = a > 0 में परिबद्ध क्षेत्रफल $e^{-a} + 4a^2 + a - 1$ है। तब अवकल समीकरण जिसका सामान्य $y = c_1 f(x) + c_2$ है, जहाँ c_1 तथा c_2 स्वैच्छिक अचर है, होगा :

(A) $\left(8e^x-1\right)\frac{d^2y}{dx^2}-\frac{dy}{dx}=0$ (B) $\left(8e^x+1\right)\frac{d^2y}{dx^2}-\frac{dy}{dx}=0$

(C) $\left(8e^x+1\right)\frac{d^2y}{dx^2}+\frac{dy}{dx}=0$ (D) $\left(8e^x-1\right)\frac{d^2y}{dx^2}+\frac{dy}{dx}=0$

Ans. C

Sol. $\int_0^a f(x)\,dx = e^{-a} + 4a^2 + a - 1$



$f(a) = -e^{-a} + 8a + 1$

$$y = C_1\left(1+8x-\frac{1}{e^x}\right)+C_2$$

$$\frac{dy}{dx}=C_1\left(8+\frac{1}{e^x}\right)\Rightarrow\frac{\left(\frac{dy}{dx}\right)}{\left(8+\frac{1}{e^x}\right)}=C_1$$

$$\frac{d^2y}{dx^2}\left(8+\frac{1}{e^x}\right)-\left(\frac{-1}{e^x}\right)\frac{dy}{dx}=0$$

113. Let y= y(x) be the solution curve of the differential equation $\sec y\frac{dy}{dx}+2x\sin y=x^3\cos y$, $y(1)=0$. Then $y\left(\sqrt{3}\right)$ is equal to :

माना अवकल समीकरा $\sec y\frac{dy}{dx}+2x\sin y=x^3\cos y$, $y(1)=0$ का हल वक्र $y=y(x)$ है। तो $y\left(\sqrt{3}\right)$ बराबर है :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{12}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{3}$

Ans. C

Sol. $\sec y\frac{dy}{dx}+2x\sin y=x^3\cos y$

$\Rightarrow \sec^2 y\frac{dy}{dx}+2x\tan y=x^3$

Let $z=\tan y$

$\frac{dz}{dx}=\sec^2 y\frac{dy}{dx}$

$\Rightarrow \frac{dz}{dx}+2xz=x^3$

I.F. $=e^{x^2}$

$\Rightarrow z.e^{x^2}=\int e^{x^2}\cdot x^3dx+c$

$\Rightarrow \tan y\cdot e^{x^2}=\frac{1}{2}\left(x^2e^{x^2}-e^{x^2}\right)+c$

$\Rightarrow \tan(0)\cdot e=\frac{1}{2}(1\cdot e-e)+c$

$\Rightarrow c=0$

$\Rightarrow \tan y=\frac{x^2-1}{2}$

$f(x)=\tan^{-1}\left(\frac{x^2-1}{2}\right)\Rightarrow f\left(\sqrt{3}\right)=\frac{\pi}{4}$

114. The solution curve, of the differential equation $2y\frac{dy}{dx}+3=5\frac{dy}{dx}$, passing through the point (0, 1) is a conic, whose vertex lies on the line :

अवकल समीकरण $2y\frac{dy}{dx}+3=5\frac{dy}{dx}$ का हल वक्र, जो (0, 1) से होकर जाता है, एक शांकव है। इस शांकव का शीर्ष निम्न में से किस रेखा पर है :

(A) $2x+3y=6$ (B) $2x+3y=9$ (C) $2x+3y=-6$ (D) $2x+3y=-9$

Ans. B

Sol. $2y\frac{dy}{dx}+3=5\frac{dy}{dx}$

$2ydy+3dx=5dy$

$y^2+3x=5y+c$, (0, 1)

$1+0=5+c$

$c=-4$

$y^2-5y=-3x-4$

$y^2-5y+\frac{25}{4}-\frac{25}{4}=3x-4$

$\left(y-\frac{5}{2}\right)^2=-3x+\frac{9}{4}$

$\left(y-\frac{5}{2}\right)^2=-3\left(x-\frac{3}{4}\right)$

$\left(\frac{3}{4},\frac{5}{2}\right)$ satisfies by $2x+3y=9$

115. The solution of the differential equation $(x^2+y^2)dx-5xy\,dy=0$, $y(1)=0$, is :

अवकल समीकरण $(x^2+y^2)dx-5xy\,dy=0$, $y(1)=0$ का हल है :

(A) $|x^2-4y^2|^5=x^2$ (B) $|x^2-2y^2|^5=x^2$ (C) $|x^2-4y^2|^6=x$ (D) $|x^2-2y^2|^6=x$

Ans. A

Sol. $(x^2+y^2)dx=5xydy$

$$\frac{dy}{dx}=\frac{x^2+y^2}{5xy}$$

Let $y=vx$

$$\frac{dy}{dx}=v+x\frac{dv}{dx}$$

$$\frac{1+v^2}{5v}=v+x\frac{dv}{dx}$$

$$x\frac{dv}{dx}=\frac{1+v^2}{5v}-v$$

$$x\frac{dv}{dx}=\frac{1-4v^2}{5v}$$

$$\frac{5v}{1-4v^2}dv=\frac{dx}{x}$$

$$\int\frac{dx}{x}=-\frac{5}{8}\int\frac{-8v}{1-4v^2}dv$$

$$\ell nx=-\frac{5}{8}\ell n\left|1-4v^2\right|+\ell nc$$

$$\ell nx+\frac{5}{8}\ell n\left|1-\frac{4y^2}{x^2}\right|=\ell nc \qquad \because\ y(1)=0$$

So $l$nc = 0

$$8\ell nx+5\ell n\left|\frac{x^2-4y^2}{x^2}\right|=0$$

$8lnx + 5ln\,|x^2 - 4y^2| - 10lnx = 0$

$ln\,|x^2 - 4y^2|^5 - lnx^2 = 0$

$|x^2 - 4y^2|^5 = x^2$

116. Let $\int_0^x\sqrt{1-\left(y'(t)\right)^2}dt=\int_0^x y(t)dt$, $0\le x\le 3$, $y\ge 0$, $y(0)=0$. Then at $x=2$, $y''+y+1$ is equal to :

माना $\int_0^x\sqrt{1-\left(y'(t)\right)^2}dt=\int_0^x y(t)dt$, $0\le x\le 3$, $y\ge 0$, $y(0)=0$ हैं। तो $x=2$ पर $y''+y+1$ बराबर है :

(A) 1 (B) $\sqrt{2}$ (C) 2 (D) $\frac{1}{2}$

Ans. A

Sol. $\sqrt{1-\left(y'(x)\right)^2} = y(x)$

$$1-\left(\frac{dy}{dx}\right)^2 = y^2$$

$$\left(\frac{dy}{dx}\right)^2 = 1-y^2$$

$$\frac{dy}{\sqrt{1-y^2}} = dx \text{ OR } \frac{dy}{\sqrt{1-y^2}} = -dx$$

$\Rightarrow \sin^{-1}y = x + c, \sin^{-1}y = -x + c$

$x = 0, y = 0 \quad \Rightarrow c = 0$

$\sin^{-1}y = x$, as $y \geq 0$

$\sin x = y$

$$\Rightarrow \frac{dy}{dx} = \cos x$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\sin x$$

$\Rightarrow -\sin x + \sin x + 1 = 1$

117. Let $x = x(t)$ and $y = y(t)$ be solutions of the differential equations $\frac{dx}{dt} + ax = 0$ and $\frac{dy}{dt} + by = 0$ respectively, a, b $\in$ R. Given that $x(0) = 2$; $y(0) = 1$ and $3y(1) = 2x(1)$, the value of t, for which $x(t) = y(t)$, is :

माना अवकल समीकरणों $\frac{dx}{dt} + ax = 0$ तथा $\frac{dy}{dt} + by = 0$, a, b $\in$ R के हल क्रमशः x = x(t) तथा y = y(t) हैं। यदि $x(0) = 2$; $y(0) = 1$ तथा $3y(1) = 2x(1)$ हैं, तो $x(t) = y(t)$ के लिए t का मान है:

(A) $\log_3 4$ (B) $\log_{\frac{2}{3}} 2$ (C) $\log_4 3$ (D) $\log_{\frac{4}{3}} 2$

Ans. D

Sol. $\frac{dx}{dt} + ax = 0$

$\Rightarrow \int \frac{1}{x} dx = -\int a dt$

$\Rightarrow \ln|x| = -at + c$

$\Rightarrow x = \pm e^c \cdot e^{-at} = \lambda e^{-at}$

$\Rightarrow x(0) = 2 \qquad \Rightarrow \lambda = 2$

$\Rightarrow x = 2e^{-at} \qquad$ _____(1)

Now, $\frac{dy}{dt} + by = 0$

$\Rightarrow \int \frac{1}{y} dy = -\int b\, dt \qquad \Rightarrow \ln(y) = -bt + c$

$\Rightarrow y = \pm e^c \cdot e^{-bt} = \mu \cdot e^{-bt}$

$y(0) = 1$

$\therefore \mu = e^{-bt}$

$3y(1) = 2x(1)$

$3e^{-b} = 2(2e^{-a}) = 4e^{-a}$

$e^{-b+a} = \frac{4}{3}$

$\Rightarrow x(t) = y(t)$

$\Rightarrow e^{-bt} = 2e^{-at}$

$\Rightarrow e^{-bt+at} = 2$

$\Rightarrow e^{(-b+a)t} = 2$

$\Rightarrow \left(\frac{4}{3}\right)^t = 2$

$\Rightarrow t\ell n\frac{4}{3} = \ell n2$

$\Rightarrow t = \log_{\frac{4}{3}} 2$

118. If $y = y(x)$ is the solution curve of the differential equation $\left(x^2 - 4\right)dy - \left(y^2 - 3y\right)dx = 0, x > 2, y\left(4\right) = \frac{3}{2}$ and the slope of the curve is never zero, then the value of $y(10)$ equals :

यदि अवकल समीकरण $\left(x^2 - 4\right)dy - \left(y^2 - 3y\right)dx = 0, x > 2, y\left(4\right) = \frac{3}{2}$ का हल वक्र $y = y(x)$ है तथा इस वक्र की प्रवणता किसी भी x के लिए शून्य नहीं है, तो $y(10)$ का मान बराबर है :

(A) $\frac{3}{1+2\sqrt{2}}$ (B) $\frac{3}{1+(8)^{1/4}}$ (C) $\frac{3}{1-2\sqrt{2}}$ (D) $\frac{3}{1-(8)^{1/4}}$

Ans. B

Sol. $(x^2-4)dy-(y^2-3y)dx=0$

$$\Rightarrow \int \frac{dy}{y^2-3y}=\int \frac{dx}{x^2-4}$$

$$\Rightarrow \frac{1}{3}\int \frac{y-(y-3)}{y(y-3)}dy=\int \frac{dx}{x^2-4}$$

$$\Rightarrow \frac{1}{3}(\ell n|y-3|-\ell n|y|)=\frac{1}{4}\ell n\left|\frac{x-2}{x+2}\right|+C$$

$$\Rightarrow \frac{1}{3}\ell n\left|\frac{y-3}{y}\right|=\frac{1}{4}\ell n\left|\frac{x-2}{x+2}\right|+C$$

At $x=4,\ y=\frac{3}{2}$

$$\therefore C=\frac{1}{4}\ell n3$$

$$\therefore \frac{1}{3}\ell n\left|\frac{y-3}{y}\right|=\frac{1}{4}\ell n\left|\frac{x-2}{x+2}\right|+\frac{1}{4}\ell n(3)$$

At $x=10$

$$\frac{1}{3}\ell n\left|\frac{y-3}{y}\right|=\frac{1}{4}\ell n\left|\frac{2}{3}\right|+\frac{1}{4}\ell n(3)$$

$$\ell n\left|\frac{y-3}{y}\right|=\ell n2^{3/4},\forall x>2,\frac{dy}{dx}<0$$

as $y(4)=\frac{3}{2}\Rightarrow y\in(0,3)$

$$-y+3=8^{1/4}\cdot y$$

$$y=\frac{3}{1+8^{1/4}}$$

119. A function $y=f(x)$ satisfies $f(x)\sin 2x+\sin x-(1+\cos^2 x)f'(x)=0$ with condition $f(0)=0$. Then, $f\left(\frac{\pi}{2}\right)$ is equal

to :

एक फलन $y = f(x)$ के लिए $f(x)\sin 2x + \sin x - (1+\cos^2 x) f'(x) = 0$ तथा $f(0) = 0$ हैं। तो $f\left(\frac{\pi}{2}\right)$ बराबर है :

(A) 2 (B) 1 (C) 0 (D) – 1

Ans. B

Sol. $\frac{dy}{dx} - \left(\frac{\sin 2x}{1+\cos^2 x}\right)y = \frac{\sin x}{1+\cos^2 x}$

$\text{If} = e^{-\int \frac{\sin 2x}{1+\cos^2 x}dx} = e^{\ln\left(1+\cos^2 x\right)} = 1+\cos^2 x$

So general solution of DE

$$y\left(1+\cos^2 x\right) = \int \frac{\sin x}{1+\cos^2 x}\cdot\left(1+\cos^2 x\right)dx$$

$$y\left(1+\cos^2 x\right) = -\cos x + c$$

$$y\left(1+\cos^2 x\right) + \cos x = c$$

were $y(0) = 0$

$0 + 1 = c$

$$y\left(1+\cos^2 x\right) + \cos x = 1$$

$$y = \frac{1-\cos x}{1+\cos^2 x}$$

$$y\left(\frac{\pi}{2}\right) = \frac{1-0}{1+0} = 1$$

120. If $\sin\left(\frac{y}{x}\right) = \log_e|x| + \frac{\alpha}{2}$ is the solution of the differential equation $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y\cos\left(\frac{y}{x}\right) + x$ and $y(1) = \frac{\pi}{3}$, then α^2 is equal to :

यदि अवकल समीकरण $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y\cos\left(\frac{y}{x}\right) + x$, $y(1) = \frac{\pi}{3}$, का हल $\sin\left(\frac{y}{x}\right) = \log_e|x| + \frac{\alpha}{2}$ है, तो

α^2 बराबर है :

(A) 3 (B) 9 (C) 12 (D) 4

Ans. A

Sol. Given differential equation is $x\cos\frac{y}{x}\frac{dy}{dx}-y\cos\frac{y}{x}=x$

$$\Rightarrow \cos\frac{y}{x}\left(x\frac{dy}{dx}-y\right)=x$$

$$\Rightarrow \cos\frac{y}{x}\left(\frac{x\frac{dy}{dx}-y\cdot 1}{x^2}\right)=\frac{1}{x}$$

Let $\frac{y}{x}=t$

$$\cos t\frac{dt}{dx}=\frac{1}{x}$$

$$\cos t\,dt=\frac{1}{x}dx$$

on integeration $\sin t=\ell n|x|+c$

$$y(1)=\frac{\pi}{3}\Rightarrow \sin\frac{\pi}{3}=0+c$$

$$\Rightarrow c=\frac{\sqrt{3}}{2}$$

$$\therefore \text{ solution is } \sin\frac{y}{x}=\ell n|x|+\frac{\sqrt{3}}{2}$$

121. Let $y=y(x)$ be the solution of the differential equation $\sec x\,dy+\{2(1-x)\tan x+x(2-x)\}dx=0$ such that $y(0)=2$. Then $y(2)$ is equal to :

माना अवकल समीकरण $\sec x\,dy+\{2(1-x)\tan x+x(2-x)\}dx=0$, $y(0)=2$ का हल $y=y(x)$ है। तो $y(2)$ बराबर है :

(A) 1 (B) $2\{\sin(2)+1\}$ (C) 2 (D) $2\{1-\sin(2)\}$

Ans. C

Sol. $$dy=\frac{2(x-1)\tan x+x(x-2)}{\sec x}dx$$

$$\Rightarrow \int dy=\int 2(x-1)\sin x+(x^2-2x)\cos x$$

$\Rightarrow y=2[(x-1)(-\cos x)-(1)(-\sin x)]+[(x^2-2x)(\sin x)-(2x-2)(-\cos x)+(2)(-\sin x)]+c$

$y(0)=2[1]+[-2]+C$

$y(0) = 2$

$C = 2$

$y(2) = 2(1 + \sin 2) - 2\sin 2 = 2$

122. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = \frac{(\tan x) + y}{\sin x(\sec x - \sin x \tan x)}$, $x \in \left(0, \frac{\pi}{2}\right)$ satisfying the condition $y\left(\frac{\pi}{4}\right) = 2$. Then $y\left(\frac{\pi}{3}\right)$, is :

माना अवकल समीकरण $\frac{dy}{dx} = \frac{(\tan x) + y}{\sin x(\sec x - \sin x \tan x)}$, $x \in \left(0, \frac{\pi}{2}\right)$ के हल $y = y(x)$ के लिए $y\left(\frac{\pi}{4}\right) = 2$ है। तो $y\left(\frac{\pi}{3}\right)$ बराबर है :

(A) $\sqrt{3}\left(2 + \log_e \sqrt{3}\right)$ (B) $\frac{\sqrt{3}}{2}\left(2 + \log_e 3\right)$ (C) $\sqrt{3}\left(2 + \log_e 3\right)$ (D) $\sqrt{3}\left(1 + 2\log_e 3\right)$

Ans. A

Sol. $\frac{dy}{dx} = \frac{\sin x + y\cos x}{\sin x \cdot \cos x\left(\frac{1}{\cos x} - \sin x \frac{\sin x}{\cos x}\right)}$

$= \frac{\sin x + y\cos x}{\sin x\left(1 - \sin^2 x\right)}$

$\frac{dy}{dx} = \sec^2 x + y \cdot 2(\text{cosec} 2x)$

$\frac{dy}{dx} - 2\text{cosec}(2x) \cdot y = \sec^2 x$

$\frac{dy}{dx} + p \cdot y = Q$

I.F. $= e^{\int p dx} = e^{\int -2\cos ec(2x)dx}$

Let $2x = t$

$2\frac{dx}{dt} = 1$

$dx = \frac{dt}{2}$

$= e^{-\int \operatorname{cosec}(t)dt}$

$= e^{-\ell n\left|\tan\frac{t}{2}\right|}$

$= e^{-\ell n|\tan x|} = \dfrac{1}{|\tan x|}$

$y(IF) = \int Q(IF)\,dx + c$

$\Rightarrow y\dfrac{1}{|\tan x|} = \int \sec^2 x\cdot\dfrac{1}{|\tan x|} + c$

$y\cdot\dfrac{1}{|\tan x|} = \int \dfrac{dt}{|t|} + c$ for $\tan x = t$

$y\cdot\dfrac{1}{|\tan x|} = \ell n|t| + c$

$y = |\tan x|\left(\ell n|\tan x| + c\right)$

Put $x = \dfrac{\pi}{4},\ y = 2$

$2 = \ell n\ 1 + c \Rightarrow c = 2$

$y = |\tan x|\ (\ell n\ |\tan x| + 2)$

$y\left(\dfrac{\pi}{3}\right) = \sqrt{3}\left(\ell n\sqrt{3} + 2\right)$

123. The solution curve of the differential equation $y\dfrac{dx}{dy} = x\left(\log_e x - \log_e y + 1\right),\ x > 0,\ y > 0$ passing through the point (e, 1) is :

बिन्दु (e, 1) से होकर जाने वाला, अवकल समीकरण $y\dfrac{dx}{dy} = x\left(\log_e x - \log_e y + 1\right),\ x > 0,\ y > 0$ का हल वक्र है :

(A) $\left|\log_e \dfrac{y}{x}\right| = y^2$ (B) $\left|\log_e \dfrac{y}{x}\right| = x$ (C) $2\left|\log_e \dfrac{x}{y}\right| = y + 1$ (D) $\left|\log_e \dfrac{x}{y}\right| = y$

Ans. D

Sol. $y\dfrac{dx}{dy} = x\left(\ell nx - \ell ny + 1\right)$

$$\frac{dx}{dy}=\frac{x}{y}\left(\ell n\frac{x}{y}+1\right)$$

Let $x=vy \Rightarrow \frac{dx}{dy}=v+y\frac{dv}{dy}$

$$\Rightarrow v+y\frac{dv}{dy}=v\ell nv+v \Rightarrow y\frac{dv}{dy}=v\ell nv \Rightarrow \int\frac{dv}{v\ell nv}=\int\frac{dy}{y}$$

$$\Rightarrow \ell n\left(\ell nv\right)=\ell ny+\ell nc \Rightarrow \ell nv=yc \Rightarrow \ell n\left(\frac{x}{y}\right)=yc$$

passes through (e, 1) $\Rightarrow \ell ne=c \Rightarrow c=1 \Rightarrow y=\ell n\left(\frac{x}{y}\right) \Rightarrow y=\left|\ell n\left(\frac{x}{y}\right)\right| \ \left(\because x>0, y>0\right)$

124. Let y = y(x) be the solution of the differential equation $\frac{dy}{dx}=2x\left(x+y\right)^3-x\left(x+y\right)-1$, y(0) = 1. Then, $\left(\frac{1}{\sqrt{2}}+y\left(\frac{1}{\sqrt{2}}\right)\right)^2$ equals :

माना अवकलन समीकरण $\frac{dy}{dx}=2x\left(x+y\right)^3-x\left(x+y\right)-1$, y(0) = 1 का हल y = y(x) है। तो $\left(\frac{1}{\sqrt{2}}+y\left(\frac{1}{\sqrt{2}}\right)\right)^2$ बराबर है :

(A) $\frac{4}{4+\sqrt{e}}$ (B) $\frac{3}{3-\sqrt{e}}$ (C) $\frac{1}{2-\sqrt{e}}$ (D) $\frac{2}{1+\sqrt{e}}$

Ans. C

Sol. $\frac{dy}{dx}=2x\left(x+y\right)^3-x\left(x+y\right)-1$

Put $x+y=t$

$$\Rightarrow \frac{dy}{dx}=\frac{dt}{dx}-1$$

$$\frac{dt}{dx}-1=2x\left(t\right)^3-xt$$

$$\Rightarrow \frac{dt}{2t^3-t}=xdx$$

$$\int \frac{1}{2t^3 - t} dt = \int x dx$$

$$\Rightarrow \int \frac{t}{2t^4 - t^2} dt = \int x dx$$

$t^2 = z$

$2tdt = dz$

$$\frac{1}{2}\int \frac{dz}{2z^2 - z} = \int x dx$$

$$\ell n \left| \frac{z - \frac{1}{2}}{z} \right| = x^2 + c$$

$$\ell n \left| \frac{(x+y)^2 - \frac{1}{2}}{(x+y)^2} \right| = x^2 + c$$

$$y(0) = 1 \Rightarrow c = \ell n\left(\frac{1}{2}\right)$$

$$\Rightarrow \frac{(x+y)^2 - \frac{1}{2}}{(x+y)^2} = e^{x^2} \times \frac{1}{2}$$

$$\Rightarrow \frac{(x+y)^2 - \frac{1}{2}}{(x+y)^2} = \sqrt{e} \times \frac{1}{2}$$

$$\Rightarrow (x+y)^2 = \frac{1}{2 - \sqrt{e}}$$

125. Let α be a non-zero real number. Suppose $f: R \to R$ is a differentiable function such that $f(0) = 2$ and $\lim\limits_{x \to -\infty} f(x) = 1$. If $f'(x) = \alpha f(x) + 3$, for all $x \in R$, then $f(-\log_e 2)$ is equal to :

माना α एक शून्येत्तर वास्तविक संख्या है। माना एक अवकलनीय फलन $f: R \to R$ के लिए $f(0) = 2$ तथा $\lim\limits_{x \to -\infty} f(x) = 1$ हैं। यदि $f'(x) = \alpha f(x) + 3$, $x \in R$ है, तो $f(-\log_e 2)$ बराबर है :

(A) 9 (B) 5 (C) 7 (D) 3

Ans. Bonus

Sol. $f(0)=2, \quad \lim\limits_{x\to-\infty} f(x)=1$

$f'(x)=\alpha f(x)+3$

$\frac{dy}{dx}=\alpha y+3$

$\int\frac{dy}{\alpha y+3}=\int dx$

$\frac{1}{\alpha}\ell n(\alpha y+3)=x+C$

$\alpha y+3=ce^{\alpha x}$

$\because \lim\limits_{x\to-\infty} f(x)=1 \qquad \Rightarrow \alpha>0$ but $\alpha=-3$

So given data contradicts the information

126. Let y = y(x) be the solution of the differential equation $(1-x^2)dy=\left[xy+(x^3+2)\sqrt{3(1-x^2)}\right]dx$, $-1<x<1, y(0)=0$. If $y\left(\frac{1}{2}\right)=\frac{m}{n}$, m and n are co-prime numbers, then m + n is equal to _____.

माना अवकल समीकरण $(1-x^2)dy=\left[xy+(x^3+2)\sqrt{3(1-x^2)}\right]dx$, $-1<x<1, y(0)=0$, का हल y = y(x) है। यदि $y\left(\frac{1}{2}\right)=\frac{m}{n}$ है, m तथा n असहभाज्य संख्याएँ हैं, तो m + n बराबर है _____।

Ans. 97

Sol. $\frac{dy}{dx}=\frac{xy+(x^3+2)\sqrt{3(1-x^2)}}{(1-x^2)}$

$\frac{dy}{dx}=\frac{xy}{1-x^2}+\frac{(x^3+2)\sqrt{3}}{\sqrt{1-x^2}}$

$\frac{dy}{dx}-\frac{xy}{1-x^2}=\frac{\sqrt{3}(x^3+2)}{\sqrt{1-x^2}}$

$$\text{I.F.} = e^{-\int \frac{x}{1-x^2}dx}$$

$$\Rightarrow e^{-\frac{1}{2}\int \frac{2x}{1-x^2}dx}$$

$$\Rightarrow e^{\frac{1}{2}\ell n\left|1-x^2\right|}$$

$$\Rightarrow e^{\ell n\sqrt{1-x^2}} \Rightarrow \sqrt{1-x^2}$$

$$\sqrt{1-x^2}\frac{dy}{dx} - \frac{\sqrt{1-x^2}}{-x^2+1}\cdot xy = \sqrt{3}\left(x^3+2\right)$$

$$\sqrt{1-x^2}\frac{dy}{dx} - \frac{1}{\sqrt{1-x^2}}\cdot xy = \sqrt{3}\left(x^3+2\right)$$

$$y\sqrt{1-x^2} = \sqrt{3}\int x^3+2dx$$

$$y\sqrt{1-x^2} = \sqrt{3}\left(\frac{x^4}{4}+2x+c\right)$$

$y(0) = 0$ $\qquad\qquad$ $\therefore\ c = 0$

$$y\sqrt{1-x^2} = \sqrt{3}\left(\frac{x^4}{4}+2x\right)$$

$$y\left(\frac{1}{2}\right) = ?$$

$$y\frac{\sqrt{3}}{2} = \sqrt{3}\left(\frac{1}{64}+1\right)$$

$$y = \frac{65}{32}$$

$$\frac{m}{n} = \frac{65}{32}$$ $\qquad\qquad$ $\therefore\ m+n = 97$

127. Let $f: R \to R$ be a thrice differentiable odd function satisfying $f'(x) \geq 0$, $f''(x) = f(x)$, $f(0) = 0$, $f'(0) = 3$. Then $9f(\log_e 3)$ is equal to _____.

माना $f: R \to R$ एक तीन बार अवकलनीय विषम फलन है, जो $f'(x) \geq 0$, $f''(x) = f(x)$, $f(0) = 0$, $f'(0) = 3$ को संतुष्ट करता है। तो $9f(\log_e 3)$ बराबर है _____.

Ans. 36

Sol. $f''(x) = f(x)$

$\Rightarrow f'(x)\, f''(x) = f'(x) \cdot f(x)$

$\Rightarrow \frac{(f'(x))^2}{2} = \frac{(f(x))^2}{2} + C$

$\Rightarrow (f'(x))^2 = (f(x))^2 + C'$

$f(0) = 0,\ f'(0) = 3 \qquad \Rightarrow C' = 9$

$\therefore (f'(x))^2 = (f(x))^2 + 9$

$f'(x) = \sqrt{(f(x))^2 + 9} \qquad \because f'(x) \geq 0$

$\int \frac{dy}{\sqrt{y^2+9}} = \int dx \Rightarrow \ell n\left|y + \sqrt{y^2+9}\right| = x + C$

$\Rightarrow f(0) = 0 \qquad \Rightarrow C = \ell n3$

$\Rightarrow y + \sqrt{y^2+9} = 3e^x$

at $x = \ell n3\ ;\ y = 4$

$\therefore 9f(\ell n3) = 36$

128. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + 2y\sec^2 x = 2\sec^2 x + 3\tan x.\sec^2 x$ such that $y(0) = \frac{5}{4}$.

Then $12\left(y\left(\frac{\pi}{4}\right) - e^{-2}\right)$ is equal to _____.

माना अवकल समीकरण $\frac{dy}{dx} + 2y\sec^2 x = 2\sec^2 x + 3\tan x.\sec^2 x,\ y(0) = \frac{5}{4}$ का हल $y = y(x)$ है। तो $12\left(y\left(\frac{\pi}{4}\right) - e^{-2}\right)$ बराबर है _____.

Ans. 21

Sol. $\frac{dy}{dx} + 2y\sec^2 x = 2\sec^2 x + 3\tan x\sec^2 x$

I.F. $= e^{\int 2\sec^2 x dx}$

I.F. $= e^{2\tan x}$

$y \cdot e^{2\tan x} = \int e^{2\tan x}(2 + 3\tan x)\sec^2 x dx$

Put $\tan x = u$

$\sec^2 x dx = du$

$$y \cdot e^{2u} = \int e^{2u}(2+3u)du$$

$$y \cdot e^{2u} \Rightarrow \frac{2e^{2u}}{2} + 3\int e^{2u} \cdot u du$$

$$y \cdot e^{2u} = e^{2u} + 3\left[\frac{ue^{2u}}{2} - \int \frac{e^{2u}}{2}\right]$$

$$ye^{2u} = e^{2u} + 3\left[\frac{ue^{2u}}{2} - \frac{e^{2u}}{4}\right] + C$$

$$ye^{2\tan x} = e^{2\tan x} + 3\left[\frac{\tan x e^{2\tan x}}{2} - \frac{e^{2\tan x}}{4}\right] + C$$

$$F(0) = \frac{5}{4}$$

$$\frac{5}{4} = 1 - \frac{3}{4} + C$$

$$\frac{5}{4} - \frac{1}{4} = C$$

$$1 = C$$

$$y = 1 + 3\left(\frac{\tan x}{2} - \frac{1}{4}\right) + 1 \cdot e^{-2\tan x}$$

$$y\left(\frac{\pi}{4}\right) = 1 + 3\left(\frac{1}{2} - \frac{1}{4}\right) + \frac{1}{e^2}$$

$$y\left(\frac{\pi}{4}\right) = \frac{7}{4} + \frac{1}{e^2}$$

$$12\left(y\left(\frac{x}{4}\right) - \frac{1}{e^2}\right) = 12\left(\frac{7}{4} + \frac{1}{e^2} - \frac{1}{e^2}\right) = 21$$

129. Let $y = f(x)$ be the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^6+4x}{\sqrt{1-x^2}}$, $-1 < x < 1$ such that $f(0) = 0$. If $6\int_{-1/2}^{1/2} f(x)dx = 2\pi - \alpha$ then α^2 is equal to _____.

माना कि $y = f(x)$, अवकल समीकरण $\frac{dy}{dx}+\frac{xy}{x^2-1}=\frac{x^6+4x}{\sqrt{1-x^2}}$, $-1 < x < 1$ का हल इस प्रकार है, कि $f(0) = 0$ है। यदि $6\int_{-1/2}^{1/2} f(x)dx = 2\pi-\alpha$ है, तो α^2 बराबर है _____।

Ans. 27

Sol. $\underbrace{\frac{dy}{dx}+\frac{x}{x^2-1}}_{P}y=\underbrace{\frac{x^6+4x}{\sqrt{1-x^2}}}_{Q}\quad -1<x<1$

$\text{I.F.} = e^{\int\frac{x}{x^2-1}dx}$

$\text{I.F.} = e^{\frac{1}{2}\int\frac{2x}{x^2-1}dx} = e^{\frac{1}{2}\ell n|x^2-1|} = \sqrt{1-x^2}$

$\Rightarrow y\times\sqrt{1-x^2}=\int\left(\frac{x^6+4x}{\sqrt{1-x^2}}\times\sqrt{1-x^2}\right)dx$

$y\times\sqrt{1-x^2}=\left(\frac{x^7}{7}+2x^2\right)+C$

$x = 0$

$y = 0$

$0 = 0 + C$

$C = 0$

$y=\frac{x^7}{7\sqrt{1-x^2}}+\frac{2x^2}{\sqrt{1-x^2}}=f(x)$

$\Rightarrow 6\int_{-\frac{1}{2}}^{\frac{1}{2}} f(x)dx = 6\times\int_{-\frac{1}{2}}^{\frac{1}{2}}\left(\underbrace{\frac{x^7}{7\sqrt{1-x^2}}}_{\text{odd}}+\underbrace{\frac{2x^2}{\sqrt{-x^2}}}_{\text{even}}\right)dx$

$\Rightarrow 6\times2\times\int_0^{\frac{1}{2}}\frac{2x^2}{\sqrt{1-x^2}}dx$

$\Rightarrow 6\times2\times\int_0^{\frac{1}{2}}\frac{2x^2}{\sqrt{1-x^2}}dx$

$$= 24\int_0^{\frac{1}{2}}\left(\frac{x^2-1+1}{\sqrt{1-x^2}}\right)dx$$

$$= 24\left(-\int_0^{\frac{1}{2}}\sqrt{1-x^2}+\int_0^{\frac{1}{2}}\frac{1}{\sqrt{1-x^2}}dx\right)$$

$$= 24\left(-\left[\frac{x}{2}\sqrt{1-x^2}+\frac{1}{2}\sin^{-1}x\right]_0^{\frac{1}{2}}+\left[\sin^{-1}x\right]_0^{\frac{1}{2}}\right)$$

$$= 24\left(-\left(\frac{\sqrt{3}}{8}+\frac{\pi}{12}-0\right)+\frac{\pi}{6}-0\right)$$

$$= 2\pi-3\sqrt{3}$$

$\alpha = 3\sqrt{3}$

$\alpha^2 = 27$

130. Let f be a differentiable function such that $2(x+2)^2 f(x) - 3(x+2)^2 = 10\int_0^x (t+2)f(t)dt$, $x \geq 0$. Then f(2) is equal to _____.

माना कि f, एक अवकलनीय फलन इस प्रकार है कि $2(x+2)^2 f(x) - 3(x+2)^2 = 10\int_0^x (t+2)f(t)dt$, $x \geq 0$ है। तब f(2) बराबर है _____.

Ans. 19

Sol. Differentiate both sides

$4(x+2) f(x) + 2(x+2)^2 f'(x) - 6(x+2) = 10(x+2) f(x)$

$2(x+2)^2 f'(x) - 6(x+2) f(x) = 6(x+2)$

$$(x+2)\frac{dy}{dx} - 3y = 3$$

$$\int\frac{dy}{dx} = 3\int\frac{dx}{x+2}$$

$\ell n(y+1) = 3\,\ell n(x+2) + C$

$(y+1) = C(x+2)^3$

$f(0)=\frac{3}{2}$

$f(2)=19$

131. Let y = y(x) be the solution of the differential equation $2\cos x\frac{dy}{dx}=\sin 2x-4y\sin x,\ x\in\left(0,\frac{\pi}{2}\right)$. If $y\left(\frac{\pi}{3}\right)=0$, then $y'\left(\frac{\pi}{4}\right)+y\left(\frac{\pi}{4}\right)$ is equal to _____.

माना y = y(x) अवकल समीकरण $2\cos x\frac{dy}{dx}=\sin 2x-4y\sin x,\ x\in\left(0,\frac{\pi}{2}\right)$ का एक हल है। यदि $y\left(\frac{\pi}{3}\right)=0$, तो $y'\left(\frac{\pi}{4}\right)+y\left(\frac{\pi}{4}\right)$ बराबर है _____.

Ans. 1

Sol. $\frac{dy}{dx}+2y\tan x=\sin x$

I.F. : $e^{\int 2\tan x dx}=\sec^2 x$

Solution of D.E.

$y\sec^2 x=\int \sec^2 x\sin x dx+c$

$y\sec^2 x=\sec x+c$

$y=\cos x+c\cos^2 x$

$y\left(\frac{\pi}{3}\right)=\frac{1}{2}+\frac{c}{4}=0 \quad \Rightarrow c=-2$

$y=\cos x-2\cos^2 x$

$y(x)=-\sin x+4\cos x\sin x$

$y'\left(\frac{\pi}{4}\right)+y\left(\frac{\pi}{4}\right)=\frac{-1}{\sqrt{2}}+4\left(\frac{1}{2}\right)+\frac{1}{\sqrt{2}}-2\left(\frac{1}{2}\right)$

$=1$

132. If y = y(x) is the solution of the differential equation, $\sqrt{4-x^2}\frac{dy}{dx}=\left(\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2-y\right)\sin^{-1}\left(\frac{x}{2}\right),\ -2\le x\le 2,$

$y(2)=\frac{\pi^2-8}{4}$, then $y^2(0)$ is equal to _____.

यदि $y = y(x)$, अवकल समीकरण, $\sqrt{4-x^2}\,\frac{dy}{dx} = \left(\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2 - y\right)\sin^{-1}\left(\frac{x}{2}\right), -2 \le x \le 2,\ y(2) = \frac{\pi^2-8}{4}$, का हल है तो $y^2(0)$ बराबर है _____।

Ans. 4

Sol. $\sqrt{4-x^2}\left(\frac{dy}{dx}\right) = \left(\sin^{-1}\left(\frac{x}{2}\right)\right)^3 - y\sin^{-1}\left(\frac{x}{2}\right)$

$$\left(\frac{dy}{dx}\right) + \frac{y\sin^{-1}\left(\frac{x}{2}\right)}{\sqrt{4-x^2}} = \frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^3}{\sqrt{4-x^2}}$$

$$\Rightarrow P(x) = \frac{\sin^{-1}\left(\frac{x}{2}\right)}{\sqrt{4-x^2}}$$

$$\Rightarrow Q(x) = \frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^3}{\sqrt{4-x^2}}$$

$$\text{I.F.} = e^{\int P(x)dx}$$

$$\int P(x)dx = \int \frac{\sin^{-1}\left(\frac{x}{2}\right)}{\sqrt{4-x^2}}dx$$

$$\sin^{-1}\left(\frac{x}{2}\right) = t$$

$$\frac{1}{\sqrt{1-\left(\frac{x}{2}\right)^2}}\left(\frac{1}{2}\right)dx = dt$$

$$\left(\frac{1}{\sqrt{4-x^2}}\right)dx = dt$$

$$\int t\,dt = \frac{t^2}{2}$$

$$e^{\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2}} = \text{I.F.}$$

$$y\cdot(\text{I.F.}) = \int Q(x)\cdot(\text{I.F.})\cdot dx$$

$$y\cdot e^{\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2}} = \int \frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^3}{\sqrt{4-x^2}}\cdot e^{\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2}}\cdot dx$$

$$\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2} = t$$

$$\frac{\sin^{-1}\left(\frac{x}{2}\right)}{\sqrt{4-x^2}}dx = dt$$

$$y\cdot e^{\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2}} = \int 2t\cdot e^t\cdot dt$$

$$y\cdot e^{\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2}} = 2\left(t\cdot e^t - e^t\right) + C$$

$$y\cdot e^{\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2}} = 2\left(\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2} - 1\right)e^{\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2}} + C$$

$$y(2) = \left(\frac{\pi^2-8}{4}\right) \Rightarrow C = 0$$

$$y \Rightarrow \left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2 - 2$$

$y^2(0) = (-2)^2 = 4$

133. Let the solution $y = y(x)$ of the differential equation $\frac{dy}{dx} - y = 1 + 4\sin x$ satisfy $y(\pi) = 1$. Then $y\left(\frac{\pi}{2}\right) + 10$ is equal to _____.

माना अवकल समीकरण $\frac{dy}{dx} - y = 1 + 4\sin x$ के हल $y = y(x)$ के लिए $y(\pi) = 1$ है। तो $y\left(\frac{\pi}{2}\right) + 10$ बराबर है _____

Ans. 7

Sol. $\frac{dy}{dx} - y = 1 + 4\sin x$ linear differential equation

I.F. $= e^{\int -dx} = e^{-x}$

So solution of linear differential equation is

$$ye^{-x} = \int e^{-x}(1 + 4\sin x)dx$$

$$ye^{-x} = -e^{-x} + 4\int e^{-x}\sin x dx$$

$$ye^{-x} = -e^{-x} + 4\frac{e^{-x}}{2}\left[-\sin x - \cos x\right] + c$$

$$y = -1 - 2\left(\sin x + \cos x\right) + ce^{x}$$

Now $y(\pi) = 1$

$\Rightarrow 1 = -1 - 2(0 - 1) + ce^{\pi}$

Hence $y = -1 - 2(\sin x + \cos x)$

$$\Rightarrow y\left(\frac{\pi}{2}\right) + 10 = -1 - 2(1 + 0) + 10$$

$= 7$

134. Let $y = y(x)$ be the solution of the differential equation $(x + y + 2)^2 dx = dy$; $y(0) = -2$. Let the maximum and minimum values of the function $y = y(x)$ in $\left[0, \frac{\pi}{3}\right]$ be α and β, respectively. If $(3\alpha + \pi)^2 + \beta^2 = \gamma + \delta\sqrt{3}$, $\gamma, \delta \in Z$, then $\gamma + \delta$ equals _____.

माना $y = y(x)$ अवकल समीकरण $(x + y + 2)^2 dx = dy$; $y(0) = -2$ का एक हल है। माना $\left[0, \frac{\pi}{3}\right]$ में फलन $y = y(x)$ के उच्चतम तथा निम्नतम मान क्रमशः α तथा β हैं। यदि $(3\alpha + \pi)^2 + \beta^2 = \gamma + \delta\sqrt{3}$, $\gamma, \delta \in Z$ तो $\gamma + \delta$ बराबर है _____

Ans. 31

Sol. $\frac{dy}{dx} = (x + y + 2)^2$

Put $x + y + 2 = t$

$1+\frac{dy}{dx}=\frac{dt}{dx}$

$\frac{dt}{dx}-1=t^2$

$\tan^{-1} t = x + c$

$\tan^{-1} (x + y + z) = x + c$

$\Rightarrow (x = 0, y = -2)$

$C = 0$

$y = \tan x - x - 2$

$\frac{dy}{dx} = \tan^2 x \geq 0$

$\forall x \in \left[0, \frac{\pi}{3}\right]$

$\Rightarrow y_{min} = y(0) = -2 = \beta$

And $y_{max} = y\left(\frac{\pi}{3}\right) = \sqrt{3} - 2 - \frac{\pi}{3} = \alpha$

Now, $(3\alpha + \pi)^2 + \beta^2$

$\left(3\sqrt{3} - 6 - \pi + \pi\right)^2 + 4$

$27 + 36 - 36\sqrt{3} + 4$

$67 - 36\sqrt{3}$

So, $\gamma = 67$

$\delta = -36$

$\Rightarrow \gamma + \delta = 31$

135. Let f be a differentiable function in the interval $(0, \infty)$ such that $f(1) = 1$ and $\lim\limits_{t\to x}\frac{t^2f(x)-x^2f(t)}{t-x}=1$ for each $x > 0$. Then $2f(2) + 3f(3)$ is equal to _____.

माना अंतराल $(0, \infty)$ में एक अवकलनीय फलन f है, जिसके लिए $f(1) = 1$ है तथा प्रत्येक $x > 0$ के लिए $\lim\limits_{t\to x}\frac{t^2f(x)-x^2f(t)}{t-x}=1$ है। तो $2f(2) + 3f(3)$ बराबर है _____

Ans. 24

Sol. $f(1) = 1$

$$\lim_{x\to\infty}\frac{t^2f(x)-x^2f(t)}{t-x}=1$$

$$\lim_{x\to\infty}\frac{2+f(x)-x^2f'(t)}{1}=1$$

$$=2xy-x^2\frac{dy}{dx}=1$$

$$=\frac{dy}{dx}-\frac{2y}{x}=\frac{-1}{x^2}$$

I.f. $e^{-\int\frac{2}{x}dx}=e^{-2\ell nx}=\frac{1}{x^2}$

$$y\,.\,\frac{1}{x^2}=\int\frac{-1}{x^4}dx$$

$$\frac{y}{x^2}=\frac{-x^{-4+1}}{-4+1}+C$$

$$\frac{y}{x^2}=\frac{1}{3x^3}+C$$

$\because f(1)=1$

$$\frac{1}{1}=\frac{1}{3}+C\ \Rightarrow\ C=1-\frac{1}{3}=\frac{2}{3}$$

$$y=\frac{1}{3x}+\frac{2}{3}x^2$$

Now $2f(2)+3f(3)=2\times\left(\frac{1}{6}+\frac{8}{3}\right)+3\times\left(\frac{1}{9}+6\right)$

$$=2\times\left(\frac{1+16}{6}\right)+3\times\left(\frac{54+1}{9/3}\right)=\frac{17}{3}+\frac{55}{3}=\frac{72}{3}=24$$ Ans.

136. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx}+\frac{2x}{\left(1+x^2\right)^2}y=xe^{\frac{1}{\left(1+x^2\right)}}$; $y(0)=0$. Then the area enclosed by the curve $f(x)=y(x)e^{-\frac{1}{\left(1+x^2\right)}}$ and the line $y-x=4$ is_____.

माना अवकल समीकरण $\frac{dy}{dx}+\frac{2x}{\left(1+x^2\right)^2}y=xe^{\frac{1}{\left(1+x^2\right)}}$; $y(0)=0$ का हल $y=y(x)$ है। तो वक्र $f(x)=y(x)e^{-\frac{1}{\left(1+x^2\right)}}$ तथा रेखा $y-x=4$ से घिरे क्षेत्र का क्षेत्रफल है_____

Ans. 18

Sol. $\frac{dy}{dx}+\frac{2x}{\left(1+x^2\right)^2}y=xe^{\frac{1}{1+x^2}}$; $y(0)=0$

I.F. of linear differential equation,

$$\text{I.F.}=e^{\int\frac{2x}{\left(1+x^2\right)^2}dx}=e^{\left(\frac{-1}{1+x^2}\right)}$$

$$\Rightarrow y\left(e^{\left(\frac{-1}{1+x^2}\right)}\right)=\int x\cdot e^{\frac{1}{1+x^2}}\cdot e^{\left(\frac{-1}{1+x^2}\right)}dx$$

$$=\frac{x^2}{2}+c$$

$\Rightarrow y(0)=0 \Rightarrow 0(e^{-1})=c \Rightarrow c=0$

$$\Rightarrow y=\frac{e^{\frac{1}{1+x^2}}\cdot x^2}{2}$$

Area between curve $ye^{\left(\frac{-1}{1+x^2}\right)}=\frac{x^2}{2}$ and $y-x=4$

$\Rightarrow 2(x+4)=x^2 \Rightarrow x^2-2x-8=0$

$\Rightarrow (x-4)(x+2)=0$

$$\int_{-2}^{4}\left[\left(x+4\right)-\frac{x^2}{2}\right]dx=\frac{x^2}{2}+4x-\frac{x^3}{6}\Bigg|_{-2}^{4}$$

$$=\left(8+16-\frac{64}{6}\right)-\left(2-8+\frac{8}{6}\right)$$

$= 30 - 12 = 18$

137. Let a conic C pass through the point (4, –2) and P(x, y), $x \geq 3$, be any point on C. Let the slope of the line touching the conic C only at a single point P be half the slope of the line joining the points P and (3, –5). If the focal distance

of the point (7, 1) on C is d, then 12d equals _____.

माना कोई शंकु C बिन्दु (4, –2) से होकर जाता है तथा C पर कोई बिन्दु P(x, y), $x \geq 3$ है। माना शंकु C को मात्र एक बिन्दु P पर स्पर्श करने वाली रेखा की प्रवणता बिन्दु P तथा (3, –5) को मिलाने वाली रेखा की प्रवणता की आधी है। यदि C पर स्थित बिन्दु (7, 1) की नाभीय दूरी d है, तो 12d बराबर है _____

Ans. 75

Sol. P(x, y) and $x \geq 3$

Slope of line at P(x, y) will be $\frac{dy}{dx} = \frac{1}{2}\left(\frac{y+5}{x-3}\right)$

$$\Rightarrow 2\frac{dy}{(y+5)} = \frac{1}{(x-3)}dx$$

$$\Rightarrow 2\ell n(y+5) = \ell n(x-3) + C$$

Passes through (4, –2)

$$\Rightarrow 2\ell n(3) = \ell n(1) + C$$

$$\Rightarrow C = 2\ell n(3)$$

$$\Rightarrow 2\ell n\left(y+5\right) = \ell n\left(x-3\right) + 2\ell n\left(3\right)$$

$$\Rightarrow 2\left(\ln\left(\frac{y+5}{3}\right)\right) = \ell n\left(x-3\right)$$

$$\Rightarrow \left(\frac{y+5}{3}\right)^2 = \left(x-3\right)$$

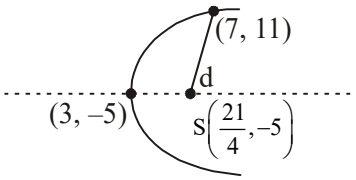
$$\Rightarrow \left(y+5\right)^2 = 9\left(x-3\right)$$

↓

Parabola

$4a = 9$

$$a = \frac{9}{4}$$



$$d = \sqrt{\left(\frac{7}{4}\right)^2 + 6^2}$$

$$d = \frac{\sqrt{625}}{4}$$

$d = \frac{25}{4}$

$12d = 75$

138. If the solution y(x) of the given differential equation $(e^y + 1)\cos x dx + e^y \sin x\, dy = 0$ passes through the point $\left(\frac{\pi}{2}, 0\right)$, then the value of $e^{y\left(\frac{\pi}{6}\right)}$ is equal to _____.

यदि अवकल समीकरण $(e^y + 1)\cos x dx + e^y \sin x\, dy = 0$ का हल y(x) बिन्दु $\left(\frac{\pi}{2}, 0\right)$ से गुजरता है, तो $e^{y\left(\frac{\pi}{6}\right)}$ का मान बराबर है _____

Ans. 3

Sol. $(e^y + 1)\cos x\, dx + e^y \sin x dy = 0$

$$\cot x dx + \frac{e^y}{e^y + 1} dy = 0$$

$$\int \cot x dx + \int \frac{e^y}{e^y + 1} dy = 0$$

$\ln|\sin x| + \ln|e^y + 1| = \ln c$

$|\sin x\,(e^y + 1)| = c$

Put & $x = \frac{\pi}{2}$ $y = 0$

$c = 2$

$\sin x\,(e^y + 1) = 2$

$e^y + 1 = 2\,\text{cosec}\, x$

$e^y = 2\,\text{cosec}\, x - 1$

$$e^{y\left(\frac{\pi}{6}\right)} = 2\,\text{cosec}\frac{\pi}{6} - 1$$

$= 2 \times 2 - 3 = 3$

139. Let $\alpha|x| = |y|e^{xy-\beta}$, $\alpha, \beta \in N$ be the solution of the differential equation $xdy - ydx + xy(xdy + ydx) = 0$, $y(1) = 2$. Then $\alpha + \beta$ is equal to _____.

माना अवकल समीकरण $xdy - ydx + xy(xdy + ydx) = 0, y(1) = 2$ का हल $\alpha|x| = |y|e^{xy-\beta}$, $\alpha, \beta \in N$ है। तो $\alpha + \beta$ बराबर है _____.

Ans. 4

Sol. $\alpha|x| = |y|\, e^{yx-\beta}$, $\alpha, \beta \in N$

$xdy - ydx + xy(xdy + ydx) = 0$

$$\frac{dy}{y} - \frac{dx}{x} + (xdy + ydx) = 0$$

$\ell n|y| - \ell n|x| + xy = c$

$y(1) = 2$

$\ell n|2| - 0 + 2 = c$

$c = 2 + \ell n2$

$\ell n|y| - \ell n|x| + xy = 2 + \ell n2$

$$\ell n|x| = \ell n\left|\frac{y}{2}\right| - 2 + xy$$

$$|x| = \left|\frac{y}{2}\right| e^{xy-2}$$

$2|x| = |y|e^{xy-2}$

$\alpha = 2$ $\beta = 2$ $\alpha + \beta = 4$

140. For a differentiable function $f: R \to R$, suppose $f'(x) = 3f(x) + \alpha$, where $\alpha \in R$, $f(0) = 1$ and $\lim\limits_{x\to-\infty} f(x) = 7$. Then $9f(-\log_e 3)$ is equal to _____.

एक अवकलनीय फलन $f: R \to R$ के लिए, माना $f'(x) = 3f(x) + \alpha$, जहाँ $\alpha \in R$, $f(0) = 1$ तथा $\lim\limits_{x\to-\infty} f(x) = 7$ हैं। तो $9f(-\log_e 3)$ बराबर है _____.

Ans. 61

Sol. $f'(x) = 3f(x) + \alpha$

$$\Rightarrow \frac{dy}{3y + \alpha} = dx$$

$$\Rightarrow \frac{1}{3}\ell n(3y + \alpha) = x + C$$

$$y(0) = 1 \Rightarrow C = \frac{1}{3}\ell n(3 + \alpha)$$

$$\frac{1}{3}\ell n\left(\frac{3y+\alpha}{3+\alpha}\right)=x$$

$$\Rightarrow y=\frac{1}{3}\left((3+\alpha)e^{3x}-\alpha\right)=f(x)$$

$$\lim_{x\to-\infty} f(x)=7\Rightarrow\alpha=-21$$

$$\Rightarrow f(x)=7-6e^{3x}$$

$$9f(-\ell n3)=61$$

141. If the solution of the differential equation $(2x + 3y – 2)\,dx + (4x + 6y – 7)\,dy = 0$, $y(0) = 3$, is $\alpha x + \beta y + 3\log_e|2x + 3y – \gamma| = 6$, then $\alpha + 2\beta + 3\gamma$ is equal to _____.

यदि अवकल समीकरण $(2x + 3y – 2)\,dx + (4x + 6y – 7)\,dy = 0$, $y(0) = 3$ का हल $\alpha x + \beta y + 3\log_e|2x + 3y – \gamma| = 6$ है, तो $\alpha + 2\beta + 3\gamma$ बराबर है _____।

Ans. 29

Sol. $(2x + 3y – 2)dx + (4x + 6y – 7)dy = 0$

Let $2x+3y=t$

$$2+3\frac{dy}{dx}=\frac{dt}{dx}$$

$$\frac{1}{3}\left[\frac{dt}{dx}-2\right]=-\left(\frac{t-2}{2t-7}\right)$$

$$\frac{dt}{dx}-2=-\left(\frac{3t-6}{2t-7}\right)$$

$$\frac{dt}{dx}=2-\frac{3t-6}{2t-7}$$

$$\frac{dt}{dx}=\frac{t-8}{2t-7}$$

$$\int\left(\frac{2t-7}{t-8}\right)dt=\int dx$$

$$=\int 2dt+\int\frac{a}{t-8}dt=\int dx$$

$$=2t+9\ln|t-8|=x+c$$

$$=2(2x+3y)+9|2x+3y-8|=x+c$$

142. If the solution curve, of the differential equation $\frac{dy}{dx}=\frac{x+y-2}{x-y}$ passing through the point (2, 1) is

$\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{\beta}\log_e\left(\alpha+\left(\frac{y-1}{x-1}\right)^2\right)=\log_e|x-1|$, then $5\beta+\alpha$ is equal to _____.

यदि अवकलन समीकरण $\frac{dy}{dx}=\frac{x+y-2}{x-y}$ का हल वक्र, जो बिंदु (2, 1) से होकर जाता है,

$\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{\beta}\log_e\left(\alpha+\left(\frac{y-1}{x-1}\right)^2\right)=\log_e|x-1|$ है, तो $5\beta+\alpha$ बराबर है _____

Ans. 11

Sol. $\frac{dy}{dx}=\frac{x+y-2}{x-y}$

$x+y-2=0$ and $x-y=0$ intersect at (1, 1)

$x=X+1, y=Y+1$

$\Rightarrow \frac{dY}{dX}=\frac{X+Y}{X-Y}$

Put $Y=tX \rightarrow \frac{dY}{dX}=t+X\frac{dt}{dX}$

$\Rightarrow t+X\frac{dt}{dX}=\frac{1+t}{1-t}$

$X\frac{dt}{dX}=\frac{1+t^2}{1-t}$

$\frac{dX}{X}=\frac{1-t}{1+t^2}dt$

$\ell n|X|=\tan^{-1}t-\frac{1}{2}\ln|1+t^2|$

$\ell n|x-1|=\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{2}\ell n\left|1+\left(\frac{y-1}{x-1}\right)^2\right|+C$

Passes through (2, 1) $\Rightarrow C=0$

$\Rightarrow \tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{2}\ell n\left(1+\left(\frac{y-1}{x-1}\right)^2\right)=\ell n|x-1|$

$\Rightarrow 5\beta + \alpha = 11$

143. If the solution curve y = y(x) of the differential equation $(1 + y^2)(1+ \log_e x)\,dx + x\,dy = 0, x > 0$ passes through the point (1, 1) and $y(e) = \dfrac{\alpha - \tan\left(\frac{3}{2}\right)}{\beta + \tan\left(\frac{3}{2}\right)}$, then $\alpha + 2\beta$ is_____.

यदि अवकल समीकरण $(1 + y^2)(1+ \log_e x)\,dx + x\,dy = 0, x > 0$ का हल वक्र $y = y(x)$, बिन्दु (1, 1) से होकर जाता है तथा $y(e) = \dfrac{\alpha - \tan\left(\frac{3}{2}\right)}{\beta + \tan\left(\frac{3}{2}\right)}$ है, तो $\alpha + 2\beta$ बराबर है_____.

Ans. 3

Sol. $\left(1+y^2\right)(1+\ell n\,x)dx + xdy = 0$

$$\left(\frac{1+\ell n\,x}{x}\right)dx = -\frac{dy}{1+y^2}$$

$$\int\left(\frac{1+\ell n\,x}{x}\right)dx = \int -\frac{dy}{1+y^2}$$

$$\tan^{-1} y = -\frac{(1+\ln x)^2}{2} + C$$

curve passes thraugh $(1, 1) \Rightarrow C = \frac{\pi}{4} + \frac{1}{2}$

$$y = \tan\left(\frac{\pi}{4} + \frac{1}{2} - \frac{(1+\ell n\,x)^2}{2}\right)$$

$$y(e) = \tan\left(\frac{\pi}{4} + \frac{1}{2} - 2\right)$$

$$y(e) = \tan\left(\frac{\pi}{4} - \frac{3}{2}\right) = \frac{1 - \tan\left(\frac{3}{2}\right)}{1 + \tan\left(\frac{3}{2}\right)}$$

Hence $\alpha = \beta = 1$

$\alpha + 2\beta = 3$

144. Let $f(x) = \sqrt{\lim_{r\to x}\left\{\frac{2r^2\left[(f(r))^2 - f(x)f(r)\right]}{r^2 - x^2} - r^3 e^{\frac{f(r)}{r}}\right\}}$ be differentiable in $(-\infty, 0) \cup (0, \infty)$ and $f(1) = 1$. Then the value of e^a, such that $f(a) = 0$, is equal to _____ .

माना $f(x) = \sqrt{\lim_{r\to x}\left\{\frac{2r^2\left[(f(r))^2 - f(x)f(r)\right]}{r^2 - x^2} - r^3 e^{\frac{f(r)}{r}}\right\}}$ अंतराल $(-\infty, 0) \cup (0, \infty)$ में अवकलनीय है तथा $f(1) = 1$ है। यदि $f(a) = 0$ है, तो e^a का मान है _____ .

Ans. 2

Sol. $f(x) = \sqrt{\lim_{r\to x}\left\{\frac{2r^2\left[(f(r))^2 - f(x)f(r)\right]}{r^2 - x^2} - r^3 e^{\frac{f(r)}{r}}\right\}}$

$\Rightarrow (f(x))^2 = xf(x)f'(x) - x^3 e^{\frac{f(x)}{x}}$ by L.H. rule

$\Rightarrow y^2 = xy\frac{dy}{dx} - x^3 e^{\frac{y}{x}}$

$\Rightarrow \frac{y}{x} = \frac{dy}{dx} - \frac{x^2}{y}e^{\frac{y}{x}}$

Let $y = vx \therefore \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\Rightarrow v = v + x\frac{dv}{dx} - \frac{x^2}{vx}e^v$

$\Rightarrow x\frac{dv}{dx} - \frac{x}{v}e^v = 0$

$\Rightarrow ve^{-v}dv - dx = 0$

$\Rightarrow (x + c)e^v + v + 1 = 0$ (Integration)

$\Rightarrow (x+c)e^{y/x}+\frac{y}{x}+1=0$

$\therefore y(1)=1 \therefore c=-1-\frac{2}{e}$

$\therefore \left(x-1-\frac{2}{e}\right)e^{y/x}+\frac{y}{x}+1=0$

$f(a)=0 \Rightarrow x=a, y=0 \therefore \left(a-1-\frac{2}{e}\right)+1=0$

$ae - e - 2 + e = 0$

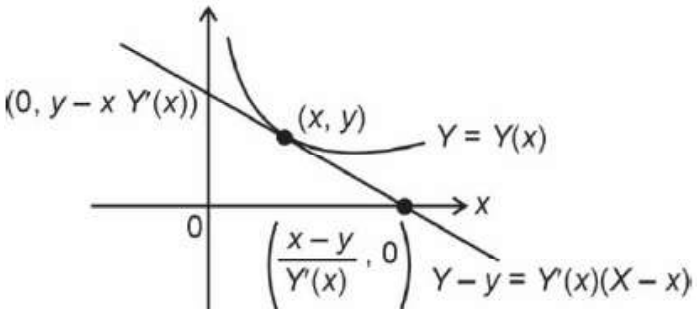
$ae = 2$

145. Let Y=Y(X) be a curve lying in the first quadrant such that the area enclosed by the line $Y-y=Y'(x)(X-x)$ and the co-ordinate axes, where (x, y) is any point on the curve, is always $\frac{-y^2}{2Y'(x)}+1$, $Y'(x)\neq 0$. If $Y(1)=1$, then $12Y(2)$ equals _____.

माना प्रथम चतुर्थांश में एक वक्र Y=Y(X) इस प्रकार है कि रेखा $Y-y=Y'(x)(X-x)$ तथा निर्देशांक अक्षों से घिरे क्षेत्र का क्षेत्रफल $\frac{-y^2}{2Y'(x)}+1$, $Y'(x)\neq 0$ है, (x, y) वक्र पर कोई बिन्दु है। यदि $Y(1)=1$ है, तो $12Y(2)$ बराबर है _____

Ans. 20

Sol.



$\Rightarrow \frac{1}{2}\left(y-xY(x)\right)\left(x-\frac{y}{Y(x)}\right)=-\frac{y^2}{2Y(x)}+1$

$\left(y-xY'(x)\right)\left(xY'(x)-y\right)=-y^2+2Y'(x)$

$yxY'(x)-y^2-x^2\left(Y'(x)\right)+xyY'(x)=-y^2+2Y'(x)$

$$\left(y-x\frac{dy}{dx}\right)\left(x\frac{dy}{dx}-y\right)=-y^2+2\frac{dy}{dx}$$

$$-y^2-x^2\left(\frac{dy}{dx}\right)^2+2xy\left(\frac{dy}{dx}\right)=-y^2+2\frac{dy}{dx}$$

$$-x^2\frac{dy}{dx}+2xy=2$$

$$\Rightarrow \frac{dy}{dx}+\left(-\frac{2}{x}\right)y=\left(-\frac{2}{x^2}\right)$$

$$\text{I.F.}=\frac{1}{x^2}$$

$$\Rightarrow \text{Solution is } \frac{y}{x^2}=\frac{-2\cdot x^{-3}}{-3}+c$$

$$\because y(1)=1$$

$$\Rightarrow c=\frac{1}{3}$$

$$\frac{y}{x^2}=\frac{2}{3x^3}+\frac{1}{3},\ \text{when } x=2,\ y=4\left(\frac{1}{12}+\frac{1}{3}\right)=\frac{5}{3}$$

$$\Rightarrow 12y(2)=20$$

146. Let $y=y(x)$ be the solution of the differential equation

$\sec^2 x\,dx+\left(e^{2y}\tan^2 x+\tan x\right)dy=0, 0<x<\frac{\pi}{2}, y\left(\frac{\pi}{4}\right)=0$. If $y\left(\frac{\pi}{6}\right)=\alpha$, then $e^{8\alpha}$ is equal to_____.

माना अवकल समीकरण $\sec^2 x\,dx+\left(e^{2y}\tan^2 x+\tan x\right)dy=0$, $0<x<\frac{\pi}{2}$, $y\left(\frac{\pi}{4}\right)=0$ का हल $y=y(x)$ है। यदि $y\left(\frac{\pi}{6}\right)=\alpha$ है, तो $e^{8\alpha}$ बराबर है_____

Ans. 9

Sol. $\sec^2 x\,dx+\left(e^{2y}\tan^2 x+\tan x\right)dy=0$

Put $\tan x=t$

$\sec^2 x\,dx=dt$

$$dt + \left(e^{2y}t^2 + t\right)dy = 0$$

$$\frac{1}{t^2} \times \frac{dt}{dy} + \frac{1}{t} = -e^{2y}$$

Put $\frac{1}{t} = u$

$$-\frac{1}{t^2}\frac{dt}{dy} = \frac{du}{dy}$$

$$-\frac{du}{dy} + u = -e^{2y}$$

$$\frac{du}{dy} - u = e^{2y}$$

I.f. $= e^{-\int dy} = e^{-y}$

$$u \cdot e^{-y} = \int e^{-y}e^{2y}dy$$

147. If x= x(t) is the solution of the differential equation $(t+1)dx = (2x + (t+1)^4)dt$, $x(0) = 2$, then $x(1)$ equals _____.

यदि अवकल समीकरण $(t + 1)dx = (2x + (t + 1)^4)dt$, $x(0) = 2$ का हल $x = x(t)$ है, तो $x(1)$ बराबर है _____

Ans. 14

Sol. $(t + 1)dx = (2x + (t + 1)^4)dt$

$$\frac{dx}{dt} - \frac{2x}{(t+1)} = (t+1)^3$$

$$IF = e^{-\int \frac{2}{t+1}dt} = \frac{1}{(t+1)^2}$$

$$\frac{x}{(t+1)^2} = \int (t+1)\,dt$$

$$\frac{x}{(t+1)^2} = \frac{t^2}{2} + t + C$$

Now $x(0) = 2$

$\Rightarrow C = 2$

$$\therefore x = \left(\frac{t^2}{2} + t + 2\right)(t+1)^2$$

$$x(1) = \left(\frac{1}{2} + 1 + 2\right)(1+1)^2$$

$$= \frac{7}{2} \times 4 = 14$$

148. If $\frac{dx}{dy} = \frac{1 + x - y^2}{y}$, $x(1) = 1$, then $5x(2)$ is equal to _____.

यदि $\frac{dx}{dy} = \frac{1 + x - y^2}{y}$, $x(1) = 1$ है, तो $5x(2)$ बराबर है _____

Ans. 5

Sol. $\frac{dx}{dy} - \frac{x}{y} = \frac{1 - y^2}{y}$

Integration factor $= e^{\int -\frac{1}{y} dy} = \frac{1}{y}$

$$x \cdot \frac{1}{y} = \int \frac{1 - y^2}{y^2} dy$$

$$\frac{x}{y} = \frac{-1}{y} - y + c$$

$$x = -1 - y^2 + cy$$

$$x(1) = 1$$

$$1 = -1 - 1 + c \Rightarrow c = 3$$

$$x = -1 - y^2 + 3y$$

$$5x(2) = 5(-1 - 4 + 6)$$

$$= 5$$